

## CAT-Kit Progress Journal

### *GitHub Version*

A copy of the progress journal I kept throughout my master's project. Details the development of CAT-Kit, the stages I took to produce this tool. Confidential files and code have been removed, this is the public version.

January 14th

To-dos:

- Confirm you can access Lugh HPC system
  - Anthony's Lugh guide: <https://maths.universityofgalway.ie/computecluster/> (Notes things we talked about like appropriate use of nodes, queue scheduling, etc.)
- Create a new GitHub repository that we will use for this project
- Read the GATK CNV pipeline documentation and attempt the tutorial:  
<https://gatk.broadinstitute.org/hc/en-us/articles/360035531092--How-to-part-I-Sensitively-detect-copy-ratio-alterations-and-allelic-segments>
- Try your hand at making a flowchart describing a software workflow.
  - Could be good to illustrate the above GATK pipeline from the tutorial as an exercise.
  - For the full project, will need to include all possible inputs and the necessary steps needed to get them into the CNV UI's input JSON format (we will talk about this more)
  - <https://app.diagrams.net/>

## Starting instructions

January 28th

First, create a file in your home directory called ".condarc". You can create this file with "nano ~/.condarc" or the text editor of your choice.

Enter all of the following:

channels:

- bioconda
- conda-forge
- r
- nodefaults

channel\_priority: strict

... and save. Then, log out of Lugh and back in. What this does is it tells Anaconda that, whenever you make an environment, these are the channels you want it to use for packages. These packages should cover most of the major things we use for bioinformatics, and notably excludes the official "default", which I had mentioned before. (Default can cost the university money, and so I think it has been IP-blocked by Anthony, which means it will produce errors.)

Once you've done this, you can load anaconda with 'module load Anaconda3'. Then, you want to create your base conda environment that you will install general-use software to. I called mine 'main', but you can call yours whatever you want, and it's fine to make several of them for different purposes.

Create the environment with 'conda create -n main'. You can also do 'conda create -n main python==3.10' if for whatever reason, you need to specify a specific older version of Python to use. (Sometimes you may find yourself wanting to use a specific piece of software that hasn't been updated and has become incompatible with newer versions.) Once this is set up, you can activate the environment with 'conda activate main' and then use it as normal.

Below is my .bashrc file. This is the file of commands that will run every time you start your system up. I recommend creating one of these the same way you created the .condarc file, except named '.bashrc'

```
# .bashrc
```

```
# Source global definitions
```

```
if [ -f /etc/bashrc ]; then
```

```
./etc/bashrc
```

```
fi
```

```
# User specific aliases and functions
```

```
export PATH=$PATH:/data4/shared/bin/links/
```

```
module load Anaconda3
```

```
conda activate main
```

A few notes on this: 'export' sends the text to the right of '=' to the system variable 'PATH'. PATH is a special variable that tells the system where to look for installed software, expecting a colon-separated list of directories. What this does is it adds my shared binaries directory, /data4/shared/bin/links to your existing path list, \$PATH. This means any binaries I link in there will be runnable for you without additional installation, which is handy. After this, the 'module load Anaconda3' and 'conda activate main' will make sure that conda is running and in your correct environment when you log in.

After you create this .bashrc file, log out and back in. If you can type 'gatk' and have it run GATK, you have probably done this correctly. To be sure, you can enter 'which gatk' and it will tell you where your GATK executable is located. It should say: /data4/shared/bin/links/gatk  
Finally, as you may have gathered from this, the shared folder for us has been created in **/data4/shared**. Currently, you should have read and execute access to this folder, meaning that you can access data and run scripts that I put in there, but would not be able to delete the files or put your own data in here. You will run any scripts you like in your own directories (which are set up for large file storage) and if there are things that should be shared or retained long term, let me know and I will copy them to the shared folder. When you want to run an analysis using files in the shared directory, you should just use the *path* to the location of the files. Do not *copy* the files, as many are very large and may have various security restrictions on them.

In the shared folder so far, you will find the following:

bin : Executables that get added to your path (currently just GATK 4.6.2)

data\_files : Data files for various cancers, etc. that you will use for your projects.\*

reference : Human reference genome files that you will use for alignment, variant calling, and so on.\*\*

\* : Some data files will be protected access, where I will have had to sign a license agreement that the data will not be used for anything outside of the agreed parameters (under penalty of revocation of any future access, lawsuits, and/or death). I will include an 'ACCESS.txt' file that tells you what the limitations of use are.

You may freely use these data to test any analyses or try out new ideas you may have.

<https://www.nature.com/articles/sdata201610>

( The linked article explains the data and there's also a manifest file with information on the individual samples)

I will soon add a folder that will be labeled ---, which will be *tightly* controlled data that can only be used for medical research. Once I have

\*\* : Reference files are things like the Human Reference Genome v38 that will be reused over and over and over again across analyses. It is preferable to keep files like this in a central location, so that we're all using the same versions and not downloading and storing a mess of data unnecessarily. If you've used reference file A for alignment, you want to make sure other analyses also use the exact same version of reference file A, otherwise things will get weird or just break. If there are reference files that you need and you're unable to find them, let me know and I will retrieve them and put them here. If you do download files and you think they should be placed here as a shared resource, let me know.



## GATK tutorial through lugh

```
salloc -n1 -N1 -p MSC
```

```
[(main) [mcollins@lugh ~]$ squeue -u mcollins
          JOBID PARTITION    NAME    USER ST       TIME  NODES NODELIST(REASON)
          10980265      MSC      bash  mcollins  R       1:24      1 compute01
[(main) [mcollins@lugh ~]$ ssh compute01
Warning: Permanently added 'compute01,192.168.11.11' (ECDSA) to the list of known hosts.
[(main) [mcollins@compute01 ~]$ module li

Currently Loaded Modules:
  1) autotools      3) gnu7/7.3.0      5) ohpc              7) java/22.0.1
  2) prun/1.3        4) openmpi3/3.1.0  6) Anaconda3/2024.02-1
```

```
(main) [mcollins@compute01 ~]$ gatk PreprocessIntervals -L targets_C.interval_list -R
/data4/shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0 --interval-merging-rule
OVERLAPPING_ONLY -O sandbox/targets_C.preprocessed.interval_list
A USER ERROR has occurred: Badly formed genome unclippedLoc: Query interval "targets_C.interval_list" is not valid for this input.
Removed -L and tried it again
```

```
(main) [mcollins@compute01 ~]$ gatk PreprocessIntervals -R
/data4/shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0 --interval-merging-rule
OVERLAPPING_ONLY -O sandbox/targets_C.preprocessed.interval_list
```

```
(main) [mcollins@compute01 ~]$ gatk CollectReadCounts -I /data4/shared/data_files/GATK_Tutorial/tutorial_11682/tumor.bam -L
sandbox/targets_C.preprocessed.interval_list --interval-merging-rule OVERLAPPING_ONLY -O sandbox/tumor.counts.hdf5
```

```
gatk --java-options "-Xmx6500m" CreateReadCountPanelOfNormals \
-I /data4/shared/data_files/GATK_Tutorial/tutorial_11682/HG00133.alt_bwamem_GRCh38DH.20150826.GBR.exome.counts.hdf5 \
-I /data4/shared/data_files/GATK_Tutorial/tutorial_11682/HG00733.alt_bwamem_GRCh38DH.20150826.PUR.exome.counts.hdf5 \
-I /data4/shared/data_files/GATK_Tutorial/tutorial_11682/NA19654.alt_bwamem_GRCh38DH.20150826.MXL.exome.counts.hdf5 \
--minimum-interval-median-percentile 5.0 \
-O sandbox/cnvponC.pon.hdf5
```

```
gatk --java-options "-Xmx12g" DenoiseReadCounts \
-I /data4/shared/data_files/GATK_Tutorial/tutorial_11682/hcc1143_T_clean.counts.hdf5 \
--count-panel-of-normals /data4/shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 \
--standardized-copy-ratios sandbox/hcc1143_T_clean.standardizedCR.tsv \
```

```
--denoised-copy-ratios sandbox/hcc1143_T_clean.denoisedCR.tsv
```

```
(main) [mcollins@compute01 sandbox]$ ls
cnvponC.pon.hdf5 hcc1143_T_clean.denoisedCR.tsv hcc1143_T_clean.standardizedCR.tsv targets_C.preprocessed.interval_list
tumor.counts.hdf5
(main) [mcollins@compute01 sandbox]$ mkdir plots
```

```
gatk PlotDenoisedCopyRatios --standardized-copy-ratios sandbox/hcc1143_T_clean.standardizedCR.tsv --denoised-copy-ratios
sandbox/hcc1143_T_clean.denoisedCR.tsv --sequence-dictionary
/data4/shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length 46709983 --output
sandbox/plots --output-prefix hcc1143_T_clean
```

A USER ERROR has occurred: Unable to execute Rscript command: Please add the Rscript directory to your environment \${PATH}

Tried to load R

```
(main) [mcollins@compute01 ~]$ module load R
(main) [mcollins@compute01 ~]$ which Rscript
/opt/ohpc/pub/libs/gnu7/R/3.5.0/bin/Rscript
```

```
gatk PlotDenoisedCopyRatios --standardized-copy-ratios sandbox/hcc1143_T_clean.standardizedCR.tsv --denoised-copy-ratios
sandbox/hcc1143_T_clean.denoisedCR.tsv --sequence-dictionary
/data4/shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length 46709983 --output
sandbox/plots --output-prefix hcc1143_T_clean
```

Using GATK jar /data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar

Running:

```
java -Dsamjdk.use_async_io_read_samtools=false -Dsamjdk.use_async_io_write_samtools=true
-Dsamjdk.use_async_io_write_tribble=false -Dsamjdk.compression_level=2 -jar
/data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar PlotDenoisedCopyRatios --standardized-copy-ratios
sandbox/hcc1143_T_clean.standardizedCR.tsv --denoised-copy-ratios sandbox/hcc1143_T_clean.denoisedCR.tsv --sequence-dictionary
/data4/shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length 46709983 --output
sandbox/plots --output-prefix hcc1143_T_clean
13:55:33.275 INFO NativeLibraryLoader - Loading libgkl_compression.so from
jar:file:/data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/com/intel/gkl/native/libgkl_compression.so
13:55:33.435 INFO PlotDenoisedCopyRatios - -----
```

```

13:55:33.439 INFO PlotDenoisedCopyRatios - The Genome Analysis Toolkit (GATK) v4.6.2.0
13:55:33.440 INFO PlotDenoisedCopyRatios - For support and documentation go to https://software.broadinstitute.org/gatk/
13:55:33.440 INFO PlotDenoisedCopyRatios - Executing as mcollins@compute01 on Linux v3.10.0-1160.88.1.el7.x86_64 amd64
13:55:33.440 INFO PlotDenoisedCopyRatios - Java runtime: Java HotSpot(TM) 64-Bit Server VM v22.0.1+8-16
13:55:33.440 INFO PlotDenoisedCopyRatios - Start Date/Time: 11 February 2026, 13:55:33 GMT
13:55:33.440 INFO PlotDenoisedCopyRatios - -----
13:55:33.440 INFO PlotDenoisedCopyRatios - -----
13:55:33.441 INFO PlotDenoisedCopyRatios - HTSJDK Version: 4.2.0
13:55:33.442 INFO PlotDenoisedCopyRatios - Picard Version: 3.4.0
13:55:33.442 INFO PlotDenoisedCopyRatios - Built for Spark Version: 3.5.0
13:55:33.445 INFO PlotDenoisedCopyRatios - HTSJDK Defaults.COMPRESSION_LEVEL : 2
13:55:33.446 INFO PlotDenoisedCopyRatios - HTSJDK Defaults.USE_ASYNC_IO_READ_FOR_SAMTOOLS : false
13:55:33.446 INFO PlotDenoisedCopyRatios - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_SAMTOOLS : true
13:55:33.446 INFO PlotDenoisedCopyRatios - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_TRIBBLE : false
13:55:33.446 INFO PlotDenoisedCopyRatios - Deflater: IntelDeflater
13:55:33.446 INFO PlotDenoisedCopyRatios - Inflater: IntelInflater
13:55:33.447 INFO PlotDenoisedCopyRatios - GCS max retries/reopens: 20
13:55:33.447 INFO PlotDenoisedCopyRatios - Requester pays: disabled
13:55:33.447 INFO PlotDenoisedCopyRatios - Initializing engine
13:55:33.447 INFO PlotDenoisedCopyRatios - Done initializing engine
13:55:33.451 INFO PlotDenoisedCopyRatios - Reading and validating input files...
13:55:34.669 INFO PlotDenoisedCopyRatios - Contigs above length threshold: {chr1=248956422, chr2=242193529, chr3=198295559,
chr4=190214555, chr5=181538259, chr6=170805979, chr7=159345973, chr8=145138636, chr9=138394717, chr10=133797422, chr11=135086622,
chr12=133275309, chr13=114364328, chr14=107043718, chr15=101991189, chr16=90338345, chr17=83257441, chr18=80373285, chr19=58617616,
chr20=64444167, chr21=46709983, chr22=50818468, chrX=156040895, chrY=57227415}
13:55:34.800 INFO PlotDenoisedCopyRatios - Writing plots to /home/mcollins/sandbox/plots...
13:55:35.101 INFO PlotDenoisedCopyRatios - Shutting down engine
[11 February 2026, 13:55:35 GMT] org.broadinstitute.hellbender.tools.copynumber.plotting.PlotDenoisedCopyRatios done. Elapsed time:
0.03 minutes.
Runtime.totalMemory()=331350016
org.broadinstitute.hellbender.utils.R.RScriptExecutorException:
Rscript exited with 1
Command Line: Rscript -e tempLibDir =
'/tmp/Rlib.17484945514140798493';source('/tmp/CNVPlottingLibrary.3041481449122730422.R');source('/tmp/PlotDenoisedCopyRatios.1346387
1478788987874.R'); --sample_name=HCC1143_tumor

```



```

--standardized_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.standardizedCR.tsv
--denoised_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.denoisedCR.tsv
--contig_names=chr1CONTIG_DELIMITERchr2CONTIG_DELIMITERchr3CONTIG_DELIMITERchr4CONTIG_DELIMITERchr5CONTIG_DELIMITERchr6CONTIG_DELIMI
TERchr7CONTIG_DELIMITERchr8CONTIG_DELIMITERchr9CONTIG_DELIMITERchr10CONTIG_DELIMITERchr11CONTIG_DELIMITERchr12CONTIG_DELIMITERchr13C
ONTIG_DELIMITERchr14CONTIG_DELIMITERchr15CONTIG_DELIMITERchr16CONTIG_DELIMITERchr17CONTIG_DELIMITERchr18CONTIG_DELIMITERchr19CONTIG_
DELIMITERchr20CONTIG_DELIMITERchr21CONTIG_DELIMITERchr22CONTIG_DELIMITERchrXCONTIG_DELIMITERchrY
--contig_lengths=248956422CONTIG_DELIMITER242193529CONTIG_DELIMITER198295559CONTIG_DELIMITER190214555CONTIG_DELIMITER181538259CONTIG
_DELIMITER170805979CONTIG_DELIMITER159345973CONTIG_DELIMITER145138636CONTIG_DELIMITER138394717CONTIG_DELIMITER133797422CONTIG_DELIMI
TER135086622CONTIG_DELIMITER133275309CONTIG_DELIMITER114364328CONTIG_DELIMITER107043718CONTIG_DELIMITER101991189CONTIG_DELIMITER9033
8345CONTIG_DELIMITER83257441CONTIG_DELIMITER80373285CONTIG_DELIMITER58617616CONTIG_DELIMITER64444167CONTIG_DELIMITER46709983CONTIG_D
ELIMITER50818468CONTIG_DELIMITER156040895CONTIG_DELIMITER57227415 --maximum_copy_ratio=4.0 --point_size_copy_ratio=0.2
--output_dir=/home/mcollins/sandbox/plots/ --output_prefix=hcc1143_T_clean
Stdout: WARNING: unknown option '--sample_name=HCC1143_tumor'
WARNING: unknown option '--standardized_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.standardizedCR.tsv'
WARNING: unknown option '--denoised_copy_ratios_file=/home/mcollins/sandbox/hcc1143_T_clean.denoisedCR.tsv'
WARNING: unknown option
'--contig_names=chr1CONTIG_DELIMITERchr2CONTIG_DELIMITERchr3CONTIG_DELIMITERchr4CONTIG_DELIMITERchr5CONTIG_DELIMITERchr6CONTIG_DELIMI
TERchr7CONTIG_DELIMITERchr8CONTIG_DELIMITERchr9CONTIG_DELIMITERchr10CONTIG_DELIMITERchr11CONTIG_DELIMITERchr12CONTIG_DELIMITERchr13
CONTIG_DELIMITERchr14CONTIG_DELIMITERchr15CONTIG_DELIMITERchr16CONTIG_DELIMITERchr17CONTIG_DELIMITERchr18CONTIG_DELIMITERchr19CONTIG
_DELIMITERchr20CONTIG_DELIMITERchr21CONTIG_DELIMITERchr22CONTIG_DELIMITERchrXCONTIG_DELIMITERchrY'
WARNING: unknown option
'--contig_lengths=248956422CONTIG_DELIMITER242193529CONTIG_DELIMITER198295559CONTIG_DELIMITER190214555CONTIG_DELIMITER181538259CONTI
G_DELIMITER170805979CONTIG_DELIMITER159345973CONTIG_DELIMITER145138636CONTIG_DELIMITER138394717CONTIG_DELIMITER133797422CONTIG_DELIM
ITER135086622CONTIG_DELIMITER133275309CONTIG_DELIMITER114364328CONTIG_DELIMITER107043718CONTIG_DELIMITER101991189CONTIG_DELIMITER903
38345CONTIG_DELIMITER83257441CONTIG_DELIMITER80373285CONTIG_DELIMITER58617616CONTIG_DELIMITER64444167CONTIG_DELIMITER46709983CONTIG_
DELIMITER50818468CONTIG_DELIMITER156040895CONTIG_DELIMITER57227415'
WARNING: unknown option '--maximum_copy_ratio=4.0'
WARNING: unknown option '--point_size_copy_ratio=0.2'
WARNING: unknown option '--output_dir=/home/mcollins/sandbox/plots/'
WARNING: unknown option '--output_prefix=hcc1143_T_clean'
Stderr: Error in library(optparse) : there is no package called 'optparse'
Calls: source -> withVisible -> eval -> eval -> library
      at org.broadinstitute.hellbender.utils.R.RScriptExecutor.getScriptException(RScriptExecutor.java:79)
      at org.broadinstitute.hellbender.utils.R.RScriptExecutor.getScriptException(RScriptExecutor.java:18)
      at org.broadinstitute.hellbender.utils.runtime.ScriptExecutor.executeCuratedArgs(ScriptExecutor.java:112)

```

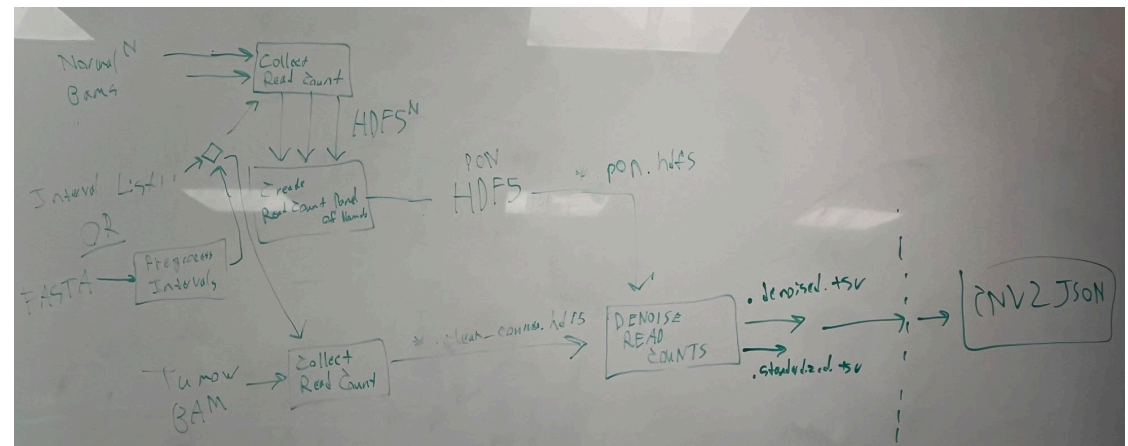
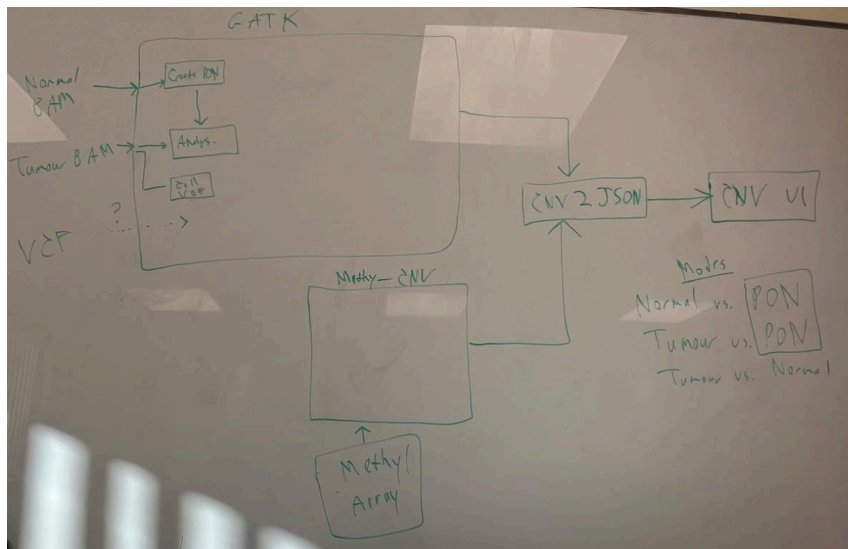
```
    at org.broadinstitute.hellbender.utils.R.RScriptExecutor.exec(RScriptExecutor.java:125)
    at
org.broadinstitute.hellbender.tools.copynumber.plotting.PlotDenoisedCopyRatios.writeDenoisingPlots(PlotDenoisedCopyRatios.java:217)
    at org.broadinstitute.hellbender.tools.copynumber.plotting.PlotDenoisedCopyRatios.doWork(PlotDenoisedCopyRatios.java:174)
    at org.broadinstitute.hellbender.cmdline.CommandLineProgram.runTool(CommandLineProgram.java:150)
    at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMainPostParseArgs(CommandLineProgram.java:203)
    at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMain(CommandLineProgram.java:222)
    at org.broadinstitute.hellbender.Main.runCommandLineProgram(Main.java:166)
    at org.broadinstitute.hellbender.Main.mainEntry(Main.java:209)
    at org.broadinstitute.hellbender.Main.main(Main.java:306)
```

## Meeting 3 notes

February 13th

### To do's

- Make flow chart
- Do GATK tutorial on --- data: 1 normal and 1 tumour, compare the plots
- Eventually make one command to create a panel of normal file (HDF5 format), using python
- Make an --- PON



### Flow Chart

- Different modes:
  - normal vs PON

- Tumour vs PON
- Tumour vs normal
- Standardised and denoised files go into JSON

### GitHub Repository

Username: meriemacollins

Password: —

Expiry: —

Two issues to make a note of on GitHub:

- A USER ERROR has occurred: Badly formed genome unclippedLoc: Query interval "targets\_C.interval\_list" is not valid for this input.  
-L  
PreprocessIntervals: More important for exome sequencing
- Stderr: Error in library(optparse) : there is no package called 'optparse'  
Need optparse R package to create plots

### Command to copy file

```
(main) [mcollins@lugh sandbox]$ cp hcc1143_T_clean.standardizedCR.tsv /data4/shared/data_files/GATK_Tutorial/tutorial_11682/sandbox/
```

### GATK on real data

- Use PoN that you already have [didn't work, made a PoN from normal blood sample]
  - Normal blood
  - Primary tumour
  - Metastatic tumour
- Use one of these instead of tumor.bam file
- Normal sample and either one of the tumours
- Tumour should look like it has copy number variants

- Make the plots

Later make an --- PON

- To get real outputs

Use python to create a PON from multiple normal bams using one command instead of 11

## GATK on real data

February 14th

```
salloc -n1 -N1 -p MSC
squeue -u mcollins
(main) [mcollins@lugh ~]$ ssh compute01
(main) [mcollins@compute01 ~]$ module load R
(main) [mcollins@compute01 ~]$ which R
/opt/ohpc/pub/libs/gnu7/R/3.5.0/bin/R
(main) [mcollins@compute01 ---]$ cd bams
(main) [mcollins@compute01 bams]$ ls
...-NB1.bam      ...-TTM1.bam      ...-TTP1.bam      cart_prj42634_202602030957.krt
...-NB1.bam.bai  ...-TTM1.bam.bai  ...-TTP1.bam.bai
[NB normal blood TP primary tumour TM metastatic tumour]
(main) [mcollins@compute01 ~]$ mkdir sandbox2

(main) [mcollins@compute01 ~]$ gatk PreprocessIntervals -R
/data4/shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0 --interval-merging-rule
OVERLAPPING_ONLY -O sandbox2/targets_C.preprocessed.interval_list

(main) [mcollins@compute01 ~] gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-TTM1.bam -L
sandbox2/targets_C.preprocessed.interval_list --interval-merging-rule OVERLAPPING_ONLY -O sandbox2/mtumor.counts.hdf5

(main) [mcollins@compute01 ~] gatk --java-options "-Xmx12g" DenoiseReadCounts -I mtumor.counts.hdf5 --count-panel-of-normals
/data4/shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 --standardized-copy-ratios mtumor.standardizedCR.tsv
--denoised-copy-ratios mtumor.denoisedCR.tsv

(main) [mcollins@compute01 sandbox2]$ gatk --java-options "-Xmx12g" DenoiseReadCounts -I mtumor.counts.hdf5 --count-panel-of-normals
/data4/shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 --standardized-copy-ratios mtumor.standardizedCR.tsv
--denoised-copy-ratios mtumor.denoisedCR.tsv
Using GATK jar /data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar
Running:
```

```

java -Dsamjdk.use_async_io_read_samtools=false -Dsamjdk.use_async_io_write_samtools=true
-Dsamjdk.use_async_io_write_tribble=false -Dsamjdk.compression_level=2 -Xmx12g -jar
/data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar DenoiseReadCounts -I mtumor.counts.hdf5 --count-panel-of-normals
/data4/shared/data_files/GATK_Tutorial/tutorial_11682/cnvponC.pon.hdf5 --standardized-copy-ratios mtumor.standardizedCR.tsv
--denoised-copy-ratios mtumor.denoisedCR.tsv
19:55:30.397 INFO NativeLibraryLoader - Loading libgkl_compression.so from
jar:file:/data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/com/intel/gkl/native/libgkl_compression.so
19:55:30.562 INFO DenoiseReadCounts - -----
19:55:30.566 INFO DenoiseReadCounts - The Genome Analysis Toolkit (GATK) v4.6.2.0
19:55:30.566 INFO DenoiseReadCounts - For support and documentation go to https://software.broadinstitute.org/gatk/
19:55:30.566 INFO DenoiseReadCounts - Executing as mcollins@compute01 on Linux v3.10.0-1160.88.1.el7.x86_64 amd64
19:55:30.566 INFO DenoiseReadCounts - Java runtime: Java HotSpot(TM) 64-Bit Server VM v22.0.1+8-16
19:55:30.567 INFO DenoiseReadCounts - Start Date/Time: 14 February 2026, 19:55:30 GMT
19:55:30.567 INFO DenoiseReadCounts - -----
19:55:30.567 INFO DenoiseReadCounts - -----
19:55:30.568 INFO DenoiseReadCounts - HTSJDK Version: 4.2.0
19:55:30.568 INFO DenoiseReadCounts - Picard Version: 3.4.0
19:55:30.568 INFO DenoiseReadCounts - Built for Spark Version: 3.5.0
19:55:30.572 INFO DenoiseReadCounts - HTSJDK Defaults.COMPRESSION_LEVEL : 2
19:55:30.573 INFO DenoiseReadCounts - HTSJDK Defaults.USE_ASYNC_IO_READ_FOR_SAMTOOLS : false
19:55:30.573 INFO DenoiseReadCounts - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_SAMTOOLS : true
19:55:30.573 INFO DenoiseReadCounts - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_TRIBBLE : false
19:55:30.574 INFO DenoiseReadCounts - Deflater: IntelDeflater
19:55:30.574 INFO DenoiseReadCounts - Inflater: IntelInflater
19:55:30.574 INFO DenoiseReadCounts - GCS max retries/reopens: 20
19:55:30.574 INFO DenoiseReadCounts - Requester pays: disabled
19:55:30.575 INFO DenoiseReadCounts - Initializing engine
19:55:30.575 INFO DenoiseReadCounts - Done initializing engine
19:55:30.578 INFO HDF5Library - Trying to load HDF5 library from:
jar:file:/data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/org/broadinstitute/hdf5/libjhdf5.2.11.0.so
19:55:30.613 INFO H5 - HDF5 library:
19:55:30.614 INFO H5 - successfully loaded.
19:55:30.620 INFO DenoiseReadCounts - Reading read-counts file (mtumor.counts.hdf5)...
19:55:30.830 INFO SVDDenoisingUtils - Validating sample intervals against original intervals used to build panel of normals...
19:55:30.884 INFO DenoiseReadCounts - Shutting down engine

```

[14 February 2026, 19:55:30 GMT] org.broadinstitute.hellbender.tools.copynumber.DenoiseReadCounts done. Elapsed time: 0.01 minutes.  
Runtime.totalMemory()=192937984

java.lang.IllegalArgumentException: Sample intervals must be identical to the original intervals used to build the panel of normals.

at org.broadinstitute.hellbender.utils.Utils.validateArg(Utils.java:798)

at org.broadinstitute.hellbender.tools.copynumber.denoising.SVDDenoisingUtils.denoise(SVDDenoisingUtils.java:119)

at

org.broadinstitute.hellbender.tools.copynumber.denoising.SVDReadCountPanelOfNormals.denoise(SVDReadCountPanelOfNormals.java:88)

at org.broadinstitute.hellbender.tools.copynumber.DenoiseReadCounts.doWork(DenoiseReadCounts.java:204)

at org.broadinstitute.hellbender.cmdline.CommandLineProgram.runTool(CommandLineProgram.java:150)

at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMainPostParseArgs(CommandLineProgram.java:203)

at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMain(CommandLineProgram.java:222)

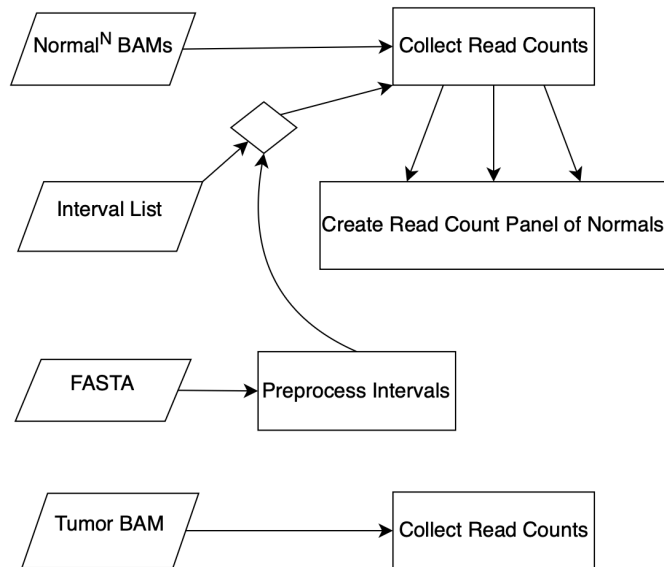
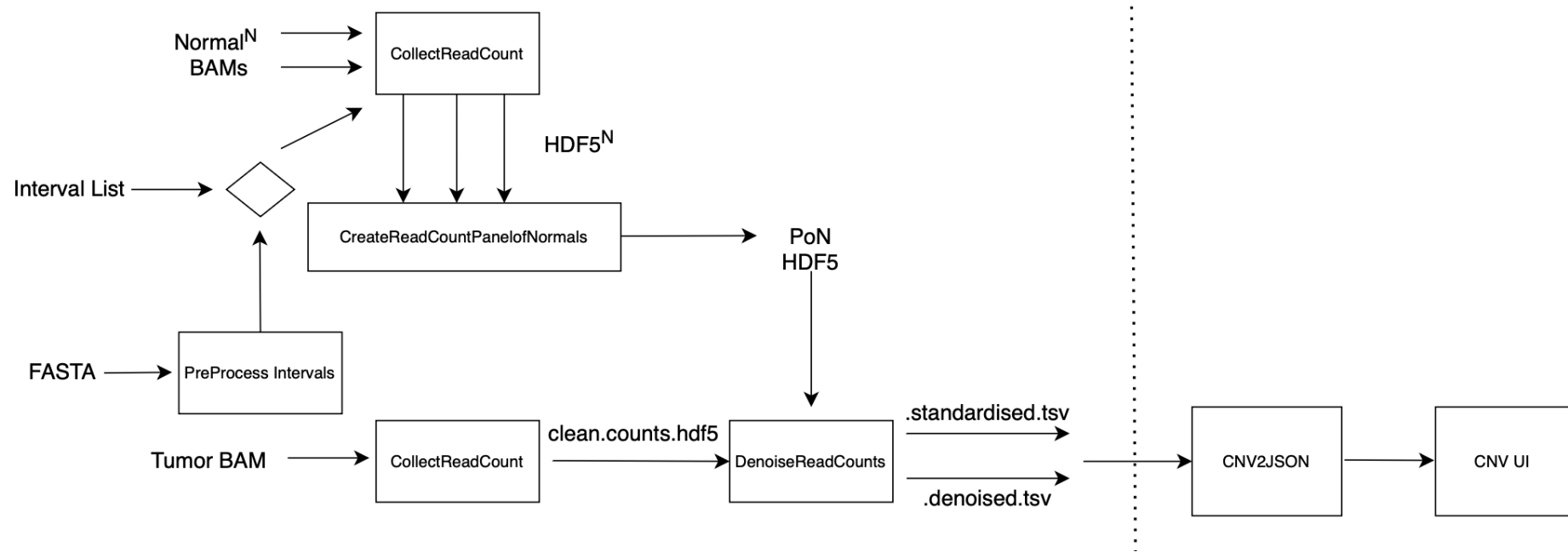
at org.broadinstitute.hellbender.Main.runCommandLineProgram(Main.java:166)

at org.broadinstitute.hellbender.Main.mainEntry(Main.java:209)

at org.broadinstitute.hellbender.Main.main(Main.java:306)



## Flow Chart Practice



Feb 17th

```
...-NB1.bam ...-TTM1.bam ...-TTP1.bam cart_prj42634_202602030957.krt
...-NB1.bam.bai ...-TTM1.bam.bai ...-TTP1.bam.bai
```

```
gatk PreprocessIntervals -R /data4/shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0
--interval-merging-rule OVERLAPPING_ONLY -O sandbox2/targets_C.preprocessed.interval_list
```

```
gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-TTM1.bam -L sandbox2/targets_C.preprocessed.interval_list
--interval-merging-rule OVERLAPPING_ONLY -O sandbox2/mtumor.counts.hdf5
```

```
gatk --java-options "-Xmx6500m" CreateReadCountPanelOfNormals -I /data4/shared/data_files/---/bams/...-NB1.bam
--minimum-interval-median-percentile 5.0 -O sandbox2/ANpon.pon.hdf5
```

#### #ERROR

java.lang.IllegalArgumentException: The collection is empty: The input header does not contain any read groups. Cannot determine a sample name.

Using GATK jar /data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar

Running:

```
java -Dsamjdk.use_async_io_read_samtools=false -Dsamjdk.use_async_io_write_samtools=true -Dsamjdk.use_async_io_write_tribble=false -Dsamjdk.compression_level=2
-Xmx6500m -jar /data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar CreateReadCountPanelOfNormals -I /data4/shared/data_files/---/bams/...-NB1.bam
--minimum-interval-median-percentile 5.0 -O sandbox2/ANpon.pon.hdf5
17:35:17.810 INFO NativeLibraryLoader - Loading libgkl_compression.so from
jar:file:/data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/com/intel/gkl/native/libgkl_compression.so
17:35:17.966 INFO CreateReadCountPanelOfNormals - -----
17:35:17.970 INFO CreateReadCountPanelOfNormals - The Genome Analysis Toolkit (GATK) v4.6.2.0
17:35:17.970 INFO CreateReadCountPanelOfNormals - For support and documentation go to https://software.broadinstitute.org/gatk/
17:35:17.970 INFO CreateReadCountPanelOfNormals - Executing as mcollins@compute02 on Linux v3.10.0-1160.88.1.el7.x86_64 amd64
17:35:17.970 INFO CreateReadCountPanelOfNormals - Java runtime: Java HotSpot(TM) 64-Bit Server VM v22.0.1+8-16
17:35:17.971 INFO CreateReadCountPanelOfNormals - Start Date/Time: 17 February 2026, 17:35:17 GMT
17:35:17.971 INFO CreateReadCountPanelOfNormals - -----
17:35:17.971 INFO CreateReadCountPanelOfNormals - -----
17:35:17.972 INFO CreateReadCountPanelOfNormals - HTSJDK Version: 4.2.0
17:35:17.972 INFO CreateReadCountPanelOfNormals - Picard Version: 3.4.0
17:35:17.972 INFO CreateReadCountPanelOfNormals - Built for Spark Version: 3.5.0
17:35:17.976 INFO CreateReadCountPanelOfNormals - HTSJDK Defaults.COMPRESSION_LEVEL : 2
17:35:17.976 INFO CreateReadCountPanelOfNormals - HTSJDK Defaults.USE_ASYNC_IO_READ_FOR_SAMTOOLS : false
17:35:17.976 INFO CreateReadCountPanelOfNormals - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_SAMTOOLS : true
17:35:17.977 INFO CreateReadCountPanelOfNormals - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_TRIBBLE : false
17:35:17.977 INFO CreateReadCountPanelOfNormals - Deflater: IntelDeflater
```

```
17:35:17.977 INFO CreateReadCountPanelOfNormals - Inflater: IntelInflater
17:35:17.977 INFO CreateReadCountPanelOfNormals - GCS max retries/reopens: 20
17:35:17.977 INFO CreateReadCountPanelOfNormals - Requester pays: disabled
17:35:17.978 INFO CreateReadCountPanelOfNormals - Initializing engine
17:35:17.978 INFO CreateReadCountPanelOfNormals - Done initializing engine
17:35:18.408 INFO SparkContext - Running Spark version 3.5.0
17:35:18.408 INFO SparkContext - OS info Linux, 3.10.0-1160.88.1.el7.x86_64, amd64
17:35:18.409 INFO SparkContext - Java version 22.0.1
17:35:18.574 WARN NativeCodeLoader - Unable to load native-hadoop library for your platform... using builtin-java classes where applicable
17:35:18.635 INFO ResourceUtils - =====
17:35:18.636 INFO ResourceUtils - No custom resources configured for spark.driver.
17:35:18.636 INFO ResourceUtils - =====
17:35:18.637 INFO SparkContext - Submitted application: CreateReadCountPanelOfNormals
17:35:18.659 INFO ResourceProfile - Default ResourceProfile created, executor resources: Map(memoryOverhead -> name: memoryOverhead, amount: 600, script: , vendor: , cores -> name: cores, amount: 1, script: , vendor: , memory -> name: memory, amount: 1024, script: , vendor: , offHeap -> name: offHeap, amount: 0, script: , vendor: ), task resources: Map(cpus -> name: cpus, amount: 1.0)
17:35:18.663 INFO ResourceProfile - Limiting resource is cpu
17:35:18.665 INFO ResourceProfileManager - Added ResourceProfile id: 0
17:35:18.722 INFO SecurityManager - Changing view acls to: mcollins
17:35:18.723 INFO SecurityManager - Changing modify acls to: mcollins
17:35:18.723 INFO SecurityManager - Changing view acls groups to:
17:35:18.724 INFO SecurityManager - Changing modify acls groups to:
17:35:18.724 INFO SecurityManager - SecurityManager: authentication disabled; ui acls disabled; users with view permissions: mcollins; groups with view permissions: EMPTY; users with modify permissions: mcollins; groups with modify permissions: EMPTY
17:35:18.994 INFO Utils - Successfully started service 'sparkDriver' on port 37272.
17:35:19.022 INFO SparkEnv - Registering MapOutputTracker
17:35:19.063 INFO SparkEnv - Registering BlockManagerMaster
17:35:19.082 INFO BlockManagerMasterEndpoint - Using org.apache.spark.storage.DefaultTopologyMapper for getting topology information
17:35:19.083 INFO BlockManagerMasterEndpoint - BlockManagerMasterEndpoint up
17:35:19.086 INFO SparkEnv - Registering BlockManagerMasterHeartbeat
17:35:19.108 INFO DiskBlockManager - Created local directory at /tmp/blockmgr-03d411c0-2867-4d4e-b994-b70163a2cfc1
17:35:19.140 INFO MemoryStore - MemoryStore started with capacity 3.6 GiB
17:35:19.155 INFO SparkEnv - Registering OutputCommitCoordinator
17:35:19.213 INFO log - Logging initialized @3678ms to org.sparkproject.jetty.util.log.Slf4jLog
17:35:19.296 INFO JettyUtils - Start Jetty 0.0.0.0:4040 for SparkUI
17:35:19.307 INFO Server - jetty-9.4.52.v20230823; built: 2023-08-23T19:29:37.669Z; git: abdcda73818a1a2c705da276edb0bf6581e7997e; jvm 22.0.1+8-16
17:35:19.325 INFO Server - Started @3791ms
17:35:19.357 INFO AbstractConnector - Started ServerConnector@7d80e1b3{HTTP/1.1, (http/1.1)}{0.0.0.0:4040}
17:35:19.357 INFO Utils - Successfully started service 'SparkUI' on port 4040.
17:35:19.377 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@552088cc{/,null,AVAILABLE,@Spark}
17:35:19.561 INFO Executor - Starting executor ID driver on host compute02.localdomain
17:35:19.561 INFO Executor - OS info Linux, 3.10.0-1160.88.1.el7.x86_64, amd64
17:35:19.562 INFO Executor - Java version 22.0.1
17:35:19.570 INFO Executor - Starting executor with user classpath (userClassPathFirst = false): ''
```

17:35:19.570 INFO Executor - Created or updated repl class loader org.apache.spark.util.MutableURLClassLoader@2f112ade for default.  
17:35:19.600 INFO Utils - Successfully started service 'org.apache.spark.network.netty.NettyBlockTransferService' on port 39098.  
17:35:19.601 INFO NettyBlockTransferService - Server created on compute02.localdomain:39098  
17:35:19.602 INFO BlockManager - Using org.apache.spark.storage.RandomBlockReplicationPolicy for block replication policy  
17:35:19.608 INFO BlockManagerMaster - Registering BlockManager BlockManagerId(driver, compute02.localdomain, 39098, None)  
17:35:19.613 INFO BlockManagerMasterEndpoint - Registering block manager compute02.localdomain:39098 with 3.6 GiB RAM, BlockManagerId(driver, compute02.localdomain, 39098, None)  
17:35:19.618 INFO BlockManagerMaster - Registered BlockManager BlockManagerId(driver, compute02.localdomain, 39098, None)  
17:35:19.619 INFO BlockManager - Initialized BlockManager: BlockManagerId(driver, compute02.localdomain, 39098, None)  
17:35:19.735 INFO ContextHandler - Stopped o.s.j.s.ServletContextHandler@552088cc{/ ,null,STOPPED,@Spark}  
17:35:19.736 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@53a16dd6{/jobs,null,AVAILABLE,@Spark}  
17:35:19.737 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@28654aff{/jobs/json,null,AVAILABLE,@Spark}  
17:35:19.738 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@5e9cb95{/jobs/job,null,AVAILABLE,@Spark}  
17:35:19.739 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@7978550b{/jobs/job/json,null,AVAILABLE,@Spark}  
17:35:19.740 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@1942a36e{/stages,null,AVAILABLE,@Spark}  
17:35:19.741 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@1f8d42d1{/stages/json,null,AVAILABLE,@Spark}  
17:35:19.742 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@22ab1b8a{/stages/stage,null,AVAILABLE,@Spark}  
17:35:19.743 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@4769378c{/stages/stage/json,null,AVAILABLE,@Spark}  
17:35:19.744 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@67671db1{/stages/pool,null,AVAILABLE,@Spark}  
17:35:19.745 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@60b4c754{/stages/pool/json,null,AVAILABLE,@Spark}  
17:35:19.746 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@735167e1{/storage,null,AVAILABLE,@Spark}  
17:35:19.747 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@21a0795f{/storage/json,null,AVAILABLE,@Spark}  
17:35:19.748 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@56d742ad{/storage/rdd,null,AVAILABLE,@Spark}  
17:35:19.749 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@5730b513{/storage/rdd/json,null,AVAILABLE,@Spark}  
17:35:19.750 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@1f4cb17b{/environment,null,AVAILABLE,@Spark}  
17:35:19.751 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@7aca299e{/environment/json,null,AVAILABLE,@Spark}  
17:35:19.752 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@6148827c{/executors,null,AVAILABLE,@Spark}  
17:35:19.753 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@66048c9e{/executors/json,null,AVAILABLE,@Spark}  
17:35:19.754 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@71458178{/executors/threadDump,null,AVAILABLE,@Spark}  
17:35:19.755 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@301e5d23{/executors/threadDump/json,null,AVAILABLE,@Spark}  
17:35:19.756 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@64a8d575{/executors/heapHistogram,null,AVAILABLE,@Spark}  
17:35:19.757 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@36cdcae0{/executors/heapHistogram/json,null,AVAILABLE,@Spark}  
17:35:19.765 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@73db7df0{/static,null,AVAILABLE,@Spark}  
17:35:19.766 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@4fce3ed7{/ ,null,AVAILABLE,@Spark}  
17:35:19.768 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@4fb99927{/api,null,AVAILABLE,@Spark}  
17:35:19.769 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@6c737fdb{/jobs/job/kill,null,AVAILABLE,@Spark}  
17:35:19.769 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@3f3f554f{/stages/stage/kill,null,AVAILABLE,@Spark}  
17:35:19.774 INFO ContextHandler - Started o.s.j.s.ServletContextHandler@254cc548{/metrics/json,null,AVAILABLE,@Spark}  
17:35:19.788 INFO CreateReadCountPanelOfNormals - Spark verbosity set to INFO (see --spark-verbosity argument)  
17:35:19.791 INFO HDF5Library - Trying to load HDF5 library from:  
    jar:file:/data4/shared/bin/gatk-4.6.2.0/gatk-package-4.6.2.0-local.jar!/org/broadinstitute/hdf5/libjhdf5.2.11.0.so  
17:35:19.825 INFO H5 - HDF5 library:  
17:35:19.825 INFO H5 - successfully loaded.

```
17:35:19.827 WARN CreateReadCountPanelOfNormals - Number of eigensamples (20) is greater than the number of input samples (1); the number of samples retained after filtering will be used instead.
17:35:19.829 INFO CreateReadCountPanelOfNormals - Retrieving intervals from first read-counts file (/data4/shared/data_files/---/bams/...-NB1.bam)...
17:35:19.883 INFO SparkContext - SparkContext is stopping with exitCode 0.
17:35:19.894 INFO AbstractConnector - Stopped Spark@7d80e1b3{HTTP/1.1, (http/1.1)}{0.0.0.0:4040}
17:35:19.900 INFO SparkUI - Stopped Spark web UI at http://compute02.localdomain:4040
17:35:19.924 INFO MapOutputTrackerMasterEndpoint - MapOutputTrackerMasterEndpoint stopped!
17:35:19.942 INFO MemoryStore - MemoryStore cleared
17:35:19.943 INFO BlockManager - BlockManager stopped
17:35:19.952 INFO BlockManagerMaster - BlockManagerMaster stopped
17:35:19.957 INFO OutputCommitCoordinator$OutputCommitCoordinatorEndpoint - OutputCommitCoordinator stopped!
17:35:19.965 INFO SparkContext - Successfully stopped SparkContext
17:35:19.965 INFO CreateReadCountPanelOfNormals - Shutting down engine
[17 February 2026, 17:35:19 GMT] org.broadinstitute.hellbender.tools.copynumber.CreateReadCountPanelOfNormals done. Elapsed time: 0.04 minutes.
Runtime.totalMemory()=125829120
java.lang.IllegalArgumentException: The collection is empty: The input header does not contain any read groups. Cannot determine a sample name.
    at org.broadinstitute.hellbender.utils.Utils.nonEmpty(Utils.java:674)
    at org.broadinstitute.hellbender.tools.copynumber.formats.metadata.MetadataUtils.readSampleName(MetadataUtils.java:23)
    at org.broadinstitute.hellbender.tools.copynumber.formats.metadata.MetadataUtils.fromHeader(MetadataUtils.java:46)
    at org.broadinstitute.hellbender.tools.copynumber.formats.collections.AbstractRecordCollection.<init>(AbstractRecordCollection.java:89)
    at org.broadinstitute.hellbender.tools.copynumber.formats.collections.AbstractLocatableCollection.<init>(AbstractLocatableCollection.java:58)
    at org.broadinstitute.hellbender.tools.copynumber.formats.collections.AbstractSampleLocatableCollection.<init>(AbstractSampleLocatableCollection.java:44)
    at org.broadinstitute.hellbender.tools.copynumber.formats.collections.SimpleCountCollection.<init>(SimpleCountCollection.java:69)
    at org.broadinstitute.hellbender.tools.copynumber.formats.collections.SimpleCountCollection.readTSV(SimpleCountCollection.java:112)
    at org.broadinstitute.hellbender.tools.copynumber.formats.collections.SimpleCountCollection.readAndSubset(SimpleCountCollection.java:99)
    at org.broadinstitute.hellbender.tools.copynumber.formats.collections.SimpleCountCollection.read(SimpleCountCollection.java:82)
    at org.broadinstitute.hellbender.tools.copynumber.CreateReadCountPanelOfNormals.runPipeline(CreateReadCountPanelOfNormals.java:268)
    at org.broadinstitute.hellbender.engine.spark.SparkCommandLineProgram.doWork(SparkCommandLineProgram.java:31)
    at org.broadinstitute.hellbender.cmdline.CommandLineProgram.runTool(CommandLineProgram.java:150)
    at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMainPostParseArgs(CommandLineProgram.java:203)
    at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMain(CommandLineProgram.java:222)
    at org.broadinstitute.hellbender.Main.runCommandLineProgram(Main.java:166)
    at org.broadinstitute.hellbender.Main.mainEntry(Main.java:209)
    at org.broadinstitute.hellbender.Main.main(Main.java:306)
17:35:19.977 INFO ShutdownHookManager - Shutdown hook called
17:35:19.978 INFO ShutdownHookManager - Deleting directory /tmp/spark-186576fc-5d17-408f-bf04-481f55310d17
```

Feb 18-21st

I am currently downloading more --- samples into the folder so that there will be more normals to make a proper PoN for subsequent use, and I will also double-check and make sure there aren't any issues with a mismatching reference genome, etc.

After getting the denoised copy ratios

- Apply ModelSegments to have GATK assemble the ratios into large blocks (going from segments of a few hundred bp to tens of millions, which makes interpretation possible).
- CNVLOH2JSON step that allows us to run the UI

GATK's CreateReadCountPanelOfNormals is expecting an HDF5 file as an input. You would need to use GATK CollectReadCounts to generate an HDF5 file from your BAM(s) and then pass those on to CreateReadCountPanelOfNormals.

This does seem like an extremely unhelpful error on GATK's part, though. If this step was expecting HDF5 input, it should not have even been able to read the BAM file and should have reported something like "file is not a valid HDF5 input" instead of something about read-groups and sample names. I did also check the BAM file for readgroup data directly, which you can do with `samtools view -H <filename> | grep '@RG'` and confirmed that readgroups are present and do contain a sample name.

## GATK on ---US files

Feb 21st

```
(main) [mcollins@compute02 bams]$ ls
```

```
...-NB1.bam          ...-TTP1.bam.bai      ...-AER6-TTM6.bam      ...-AFE4-TTM4.bam.bai
...-NB1.bam.bai      ...-AER6-TTM10.bam    ...-AER6-TTM6.bam.bai  ...-AFE4-TTM6.bam
...-TTM1.bam         ...-AER6-TTM10.bam.bai ...-AER6-TTM9.bam      ...-AFE4-TTM6.bam.bai
...-TTM1.bam.bai     ...-AER6-TTM2.bam     ...-AER6-TTM9.bam.bai  ---_downloader.py
...-TTP1.bam         ...-AER6-TTM2.bam.bai ...-AFE4-TTM4.bam      temp-000000
```

```
gatk PreprocessIntervals -R /data4/shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz --bin-length 0
--interval-merging-rule OVERLAPPING_ONLY -O sandbox2/targets_C.preprocessed.interval_list
```

```
#READ COUNTS FOR ...-NB1
```

```
gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-NB1.bam -L sandbox2/targets_C.preprocessed.interval_list
--interval-merging-rule OVERLAPPING_ONLY -O sandbox2/Anorm.counts.hdf5
```

```
#PoN MADE FROM ...-NB1
```

```
gatk --java-options "-Xmx6500m" CreateReadCountPanelOfNormals -I sandbox2/Anorm.counts.hdf5 --minimum-interval-median-percentile 5.0
-O sandbox2/ANpon.pon.hdf5
```

```
#...-NB1 TSV FILES
```

```
gatk --java-options "-Xmx12g" DenoiseReadCounts \
-I sandbox2/Anorm.counts.hdf5 \
--count-panel-of-normals sandbox2/ANpon.pon.hdf5 \
--standardized-copy-ratios sandbox2/...NB1.standardizedCR.tsv \
--denoised-copy-ratios sandbox2/...NB1.denoisedCR.tsv
```

```
#TRIED TO PLOT GRAPHS, REQUIRES OPTPARSE PACKAGE
```

```
gatk PlotDenoisedCopyRatios --standardized-copy-ratios ...NB1.standardizedCR.tsv --denoised-copy-ratios ...NB1.denoisedCR.tsv
--sequence-dictionary /data4/shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length
46709983 --output plots --output-prefix ...NB1
```

```
Stderr: Error in library(optparse) : there is no package called 'optparse'
```

```
#Installed optparse
```

```
module avail R
```

```
module load R/R-4.0.2
```

```
R
```

```
install.packages("optparse", lib=~ /R/library")
```

```
q()
```

```
#REQUIRES DATA.TABLE R PACKAGE
```

```
gatk PlotDenoisedCopyRatios --standardized-copy-ratios ...NB1.standardizedCR.tsv --denoised-copy-ratios ...NB1.denoisedCR.tsv
```

```
--sequence-dictionary /data4/shared/data_files/GATK_Tutorial/tutorial_11682/Homo_sapiens_assembly38.dict --minimum-contig-length
```

```
46709983 --output plots --output-prefix ...NB1
```

```
Stderr: Error in library(data.table) : there is no package called 'data.table'
```

```
#Attempted to install data.table
```

```
module avail R
```

```
module load R/R-4.0.2
```

```
R
```

```
install.packages("data.table", lib=~ /R/library")
```

```
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/bin/exec/R: /opt/ohpc/pub/apps/gcc/10.3.1/lib64/libgomp.so.1: no version information available (required by  
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/lib/libR.so)
```

```
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/bin/exec/R: /opt/ohpc/pub/apps/gcc/10.3.1/lib64/libgomp.so.1: no version information available (required by  
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/lib/libR.so)
```

```
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/bin/exec/R: /opt/ohpc/pub/apps/gcc/10.3.1/lib64/libstdc++.so.6: no version information available (required by  
/usr/lib64/libicuuc.so.50)
```

```
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/bin/exec/R: /opt/ohpc/pub/apps/gcc/10.3.1/lib64/libstdc++.so.6: no version information available (required by  
/usr/lib64/libicuuc.so.50)
```

```
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/bin/exec/R: /opt/ohpc/pub/apps/gcc/10.3.1/lib64/libstdc++.so.6: no version information available (required by  
/usr/lib64/libicui18n.so.50)
```

```
/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/bin/exec/R: /opt/ohpc/pub/apps/gcc/10.3.1/lib64/libstdc++.so.6: no version information available (required by  
/usr/lib64/libicui18n.so.50)
```

```
* installing *source* package 'data.table' ...
```

```
** package 'data.table' successfully unpacked and MD5 sums checked
```

```
** using staged installation
```

```
gcc 10.3.1
```

```
zlib 1.2.7 is available ok
```

```
* checking if R installation supports OpenMP without any extra hints... no
```



```

* checking if R installation supports openmp with "-fopenmp" flag... no
***
*** OpenMP not supported! data.table uses OpenMP to automatically
*** parallelize operations like sorting, grouping, file reading, etc.
*** For details on how to install the necessary toolchains on your OS see:
*** https://github.com/Rdatatable/data.table/wiki/Installation
*** Continuing installation without OpenMP support...
***
** libs
gcc -I"/opt/ohpc/pub/apps/R/R-4.0.2/lib64/R/include" -DNDEBUG -I/usr/local/include -fvisibility=hidden -fpic -g -O2 -c assign.c -o assign.o
/opt/ohpc/pub/apps/gcc/10.3.1/bin/../libexec/gcc/x86_64-pc-linux-gnu/10.3.1/cc1: error while loading shared libraries: libmpfr.so.6: cannot open shared object file:
No such file or directory
make: *** [assign.o] Error 1
ERROR: compilation failed for package 'data.table'
* removing '/home/mcollins/R/library/data.table'
The downloaded source packages are in
  '/tmp/RtmpHfosJh/downloaded_packages'
Warning message:
In install.packages("data.table", lib = "~/R/library") :
  installation of package 'data.table' had non-zero exit status

#READ COUNTS FOR ...-TTM1.bam
gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-TTM1.bam -L sandbox2/targets_C.preprocessed.interval_list
--interval-merging-rule OVERLAPPING_ONLY -O sandbox2/...TTM1.counts.hdf5

#...-TTM1 TSV FILES USING PoN MADE FROM ...-NB1
gatk --java-options "-Xmx12g" DenoiseReadCounts \
  -I sandbox2/...TTM1.counts.hdf5 \
  --count-panel-of-normals sandbox2/ANpon.pon.hdf5 \
  --standardized-copy-ratios sandbox2/...TTM1.standardizedCR.tsv \
  --denoised-copy-ratios sandbox2/...TTM1.denoisedCR.tsv

#Normal blood allelic counts ...-NB1
gatk --java-options "-Xmx3g" CollectAllelicCounts \
  -L targets_C.preprocessed.interval_list \
  -I /data4/shared/data_files/---/bams/...-NB1.bam \
  -R /data4/shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz \
  -O ...NB1.allelicCounts.tsv

```

```
#Metastatic tumour allelic counts ...-TTM1
gatk --java-options "-Xmx3g" CollectAllelicCounts \
  -L targets_C.preprocessed.interval_list \
  -I /data4/shared/data_files/---/bams/...-TTM1.bam \
  -R /data4/shared/reference/Broad/Homo_sapiens_assembly38_noALT_noHLA_noDecoy.fasta.gz \
  -O sandbox2/A...TTM1.allelicCounts.tsv
```

A USER ERROR has occurred: Badly formed genome unclippedLoc: Query interval "targets\_C.preprocessed.interval\_list" is not valid for this input.

Tried without -L

A USER ERROR has occurred: Argument intervals was missing: Argument 'intervals' is required

Unsure which interval to use

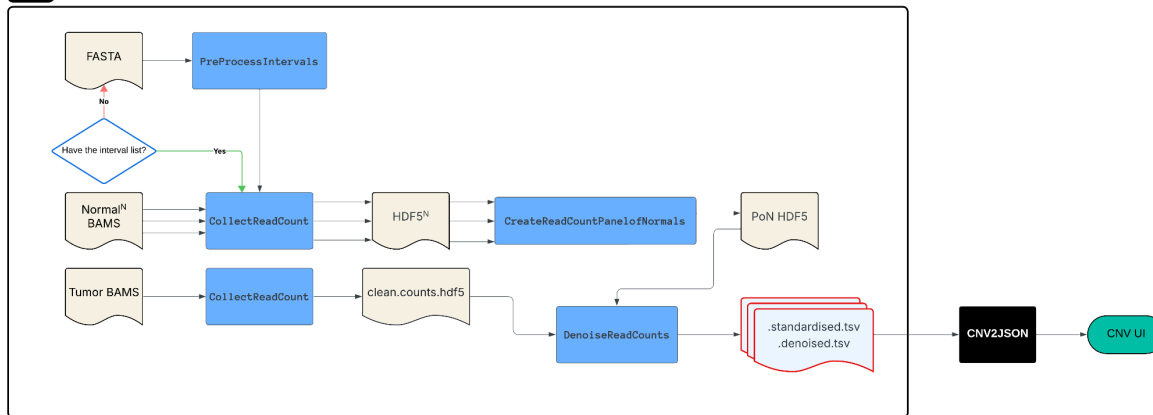
```
#Model segments
gatk --java-options "-Xmx4g" ModelSegments \
  --denoised-copy-ratios ...TTM1.denoisedCR.tsv \
  --allelic-counts A...TTM1.allelicCounts.tsv \
  --normal-allelic-counts ...NB1.allelicCounts.tsv \
  --output sandbox2 \
  --output-prefix ...TTM1_clean

gatk CallCopyRatioSegments \
  --input ...TTM1_clean.cr.seg \
  --output sandbox2/hcc1143_T_clean.called.seg
```

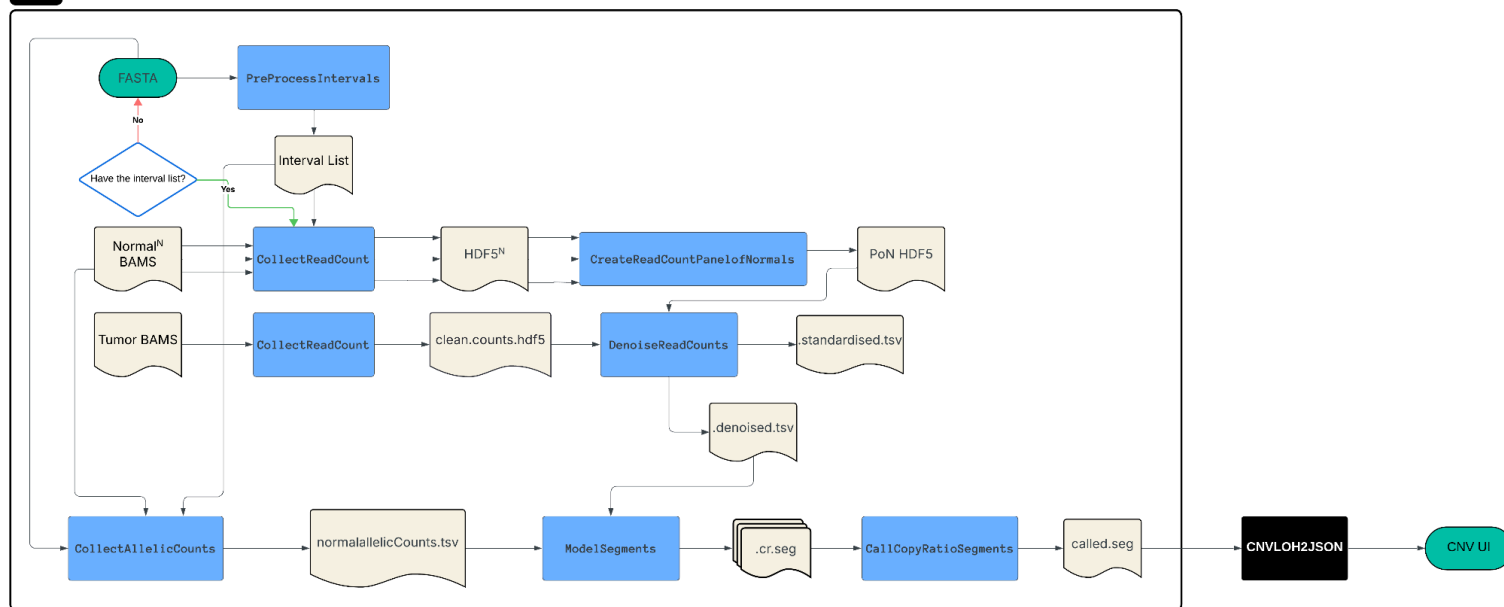
---.Exome\_2019.bed

## Flowchart practice

GATK



GATK



## Incorrect reference file used for CollectAllelicCounts

```
gatk --java-options "-Xmx3g" CollectAllelicCounts \
-L /data4/shared/reference/bed_files/---.Exome_2019.bed \
-I /data4/shared/data_files/---/bams/...-...-TTM1.bam \
-R /data4/shared/reference/HG38/genome.fa \
-O sandbox2/A...TTM1.allelicCounts.tsv
```

\*\*\*\*\*

A USER ERROR has occurred: Badly formed genome unclippedLoc: Contig chr1\_KI270766v1\_alt given as location, but this contig isn't present in the Fasta sequence dictionary

\*\*\*\*\*

\*\*\*\*\*

A USER ERROR has occurred: Fasta dict file file:///data4/shared/reference/HG38/genome.dict for reference file:///data4/shared/reference/HG38/genome.fa does not exist. Please see <https://gatk.broadinstitute.org/hc/articles/360035531652-FASTA-Reference-genome-format> for help creating it.

\*\*\*\*\*

Solution: [genome.fa](#) is the wrong reference file. hs38DH must be used instead.

## Correct Reference file for allelic counts

hs38DH = reference file used for aligning the --- exome data

#Metastatic tumour allelic counts ...-TTM1

```
gatk --java-options "-Xmx3g" CollectAllelicCounts \
-L /data4/shared/reference/bed_files/---.Exome_2019.bed \
-I /data4/shared/data_files/---/bams/...-...-TTM1.bam \
-R /data4/shared/reference/hs38DH/hs38DH.fa \
-O sandbox2/A...TTM1.allelicCounts.tsv
```

[3 March 2026, 13:40:30 GMT] org.broadinstitute.hellbender.tools.copynumber.CollectAllelicCounts done. Elapsed time: 80.71 minutes.

Runtime.totalMemory()=3221225472

java.lang.OutOfMemoryError: Java heap space: failed reallocation of scalar replaced objects

Will rerun the tutorial from the beginning with the new reference file (hs38DH)

```
gatk PreprocessIntervals -R /data4/shared/reference/hs38DH/hs38DH.fa --bin-length 0 --interval-merging-rule OVERLAPPING_ONLY -O
sandbox3/targets_C.preprocessed.interval_list
```

#READ COUNTS FOR ...-NB1

```
gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-...-NB1.bam -L sandbox3/targets_C.preprocessed.interval_list
--interval-merging-rule OVERLAPPING_ONLY -O sandbox3/Anorm.counts.hdf5
```

#READ COUNTS FOR ...-TTM1.bam

```
gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-...-TTM1.bam -L sandbox3/targets_C.preprocessed.interval_list
--interval-merging-rule OVERLAPPING_ONLY -O sandbox3/...TTM1.counts.hdf5
```

#PoN MADE FROM ...-NB1

```
gatk --java-options "-Xmx6500m" CreateReadCountPanelOfNormals -I sandbox3/Anorm.counts.hdf5 --minimum-interval-median-percentile 5.0
-O sandbox3/ANpon.pon.hdf5
```

#Metastatic tumour allelic counts ...-TTM1

```
gatk --java-options "-Xmx3g" CollectAllelicCounts \
-L /data4/shared/reference/bed_files/---.Exome_2019.bed \
-I /data4/shared/data_files/---/bams/...-...-TTM1.bam \
-R /data4/shared/reference/hs38DH/hs38DH.fa \
-O A...TTM1.allelicCounts.tsv
```

14:11:41.072 INFO CollectAllelicCounts - Shutting down engine

[8 March 2026, 14:11:41 GMT] org.broadinstitute.hellbender.tools.copynumber.CollectAllelicCounts done. Elapsed time: 80.94 minutes.

Runtime.totalMemory()=3221225472

java.lang.OutOfMemoryError: Java heap space: failed reallocation of scalar replaced objects

at org.broadinstitute.hellbender.utils.locusiterator.AlignmentStateMachine.makePileupElement(AlignmentStateMachine.java:308)

at org.broadinstitute.hellbender.utils.locusiterator.LocusIteratorByState.lazyLoadNextAlignmentContext(LocusIteratorByState.java:316)

at org.broadinstitute.hellbender.utils.locusiterator.LocusIteratorByState.hasNext(LocusIteratorByState.java:224)

at org.broadinstitute.hellbender.utils.locusiterator.IntervalAlignmentContextIterator.advanceAlignmentContext(IntervalAlignmentContextIterator.java:104)

at

org.broadinstitute.hellbender.utils.locusiterator.IntervalAlignmentContextIterator.advanceAlignmentContextToCurrentInterval(IntervalAlignmentContextIterator.java:99)

at org.broadinstitute.hellbender.utils.locusiterator.IntervalAlignmentContextIterator.next(IntervalAlignmentContextIterator.java:69)

at org.broadinstitute.hellbender.utils.locusiterator.IntervalAlignmentContextIterator.next(IntervalAlignmentContextIterator.java:21)

```
at java.base/java.util.Iterator.forEachRemaining(Iterator.java:133)
at org.broadinstitute.hellbender.engine.LocusWalker.traverse(LocusWalker.java:152)
at org.broadinstitute.hellbender.engine.GATKTool.doWork(GATKTool.java:1119)
at org.broadinstitute.hellbender.cmdline.CommandLineProgram.runTool(CommandLineProgram.java:150)
at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMainPostParseArgs(CommandLineProgram.java:203)
at org.broadinstitute.hellbender.cmdline.CommandLineProgram.instanceMain(CommandLineProgram.java:222)
at org.broadinstitute.hellbender.Main.runCommandLineProgram(Main.java:166)
at org.broadinstitute.hellbender.Main.mainEntry(Main.java:209)
at org.broadinstitute.hellbender.Main.main(Main.java:306)
```

#Normal blood allelic counts ...-NB1

```
gatk --java-options "-Xmx3g" CollectAllelicCounts \
  -L targets_C.preprocessed.interval_list \
  -I /data4/shared/data_files/---/bams/...-...-NB1.bam \
  -R /data4/shared/reference/hs38DH/hs38DH.fa \
  -O ...NB1.allelicCounts.tsv
```

#Model segments

```
gatk --java-options "-Xmx4g" ModelSegments \
  --denoised-copy-ratios ...TTM1.denoisedCR.tsv \
  --allelic-counts A...TTM1.allelicCounts.tsv \
  --normal-allelic-counts ...NB1.allelicCounts.tsv \
  --output sandbox3 \
  --output-prefix ...TTM1_clean
```

Solution: Out of memory error. Use sbatch for more space.

Used sbatch for more memory

Solves memory error

```
#Metastatic tumour allelic counts ...-TTM1
```

```
gatk --java-options "-Xmx15g" CollectAllelicCounts \  
-L /data4/shared/reference/bed_files/---.Exome_2019.bed \  
-I /data4/shared/data_files/---/bams/...-...-TTM1.bam \  
-R /data4/shared/reference/hs38DH/hs38DH.fa \  
-O A...TTM1.allelicCounts.tsv
```

```
#Normal blood allelic counts ...-NB1
```

```
gatk --java-options "-Xmx15g" CollectAllelicCounts \  
-L /data4/shared/reference/bed_files/---.Exome_2019.bed \  
-I /data4/shared/data_files/---/bams/...-...-NB1.bam \  
-R /data4/shared/reference/hs38DH/hs38DH.fa \  
-O ...NB1.allelicCounts.tsv
```

```
#SBATCH --job-name=GATK2      # create a short name for your job
```

```
#SBATCH --nodes=1            # Number of nodes for this job
```

```
#SBATCH --ntasks=1          # total number of tasks across all nodes
```

```
#SBATCH --cpus-per-task=1    # cpu-cores per task (>1 if multi-threaded tasks)
```

```
#SBATCH --mem=16G            # memory to allocate per compute node.
```

```
#SBATCH --time=48:00:00      # total run time limit (HH:MM:SS)
```

```
#SBATCH --mail-type=ALL      # send email for all events
```

```
#SBATCH --mail-user=m.collins40@universityofgalway.ie
```

```
gatk --java-options "-Xmx15g" DenoiseReadCounts \  
-I sandbox3/...TTM1.counts.hdf5 \  
--count-panel-of-normals sandbox3/ANpon.pon.hdf5 \  
--standardized-copy-ratios sandbox3/...TTM1.standardizedCR.tsv \  
--denoised-copy-ratios sandbox3/...TTM1.denoisedCR.tsv
```

```
#SBATCH --job-name=GATK2      # create a short name for your job
```

```
#SBATCH --nodes=1            # Number of nodes for this job
```

```
#SBATCH --ntasks=1          # total number of tasks across all nodes
```

```
#SBATCH --cpus-per-task=1    # cpu-cores per task (>1 if multi-threaded tasks)
#SBATCH --mem=32G           # memory to allocate per compute node.
#SBATCH --time=48:00:00     # total run time limit (HH:MM:SS)
#SBATCH --mail-type=ALL     # send email for all events
#SBATCH --mail-user=m.collins40@universityofgalway.ie
```

```
gatk --java-options "-Xmx31g" ModelSegments \
  --denoised-copy-ratios sandbox3/...TTM1.denoisedCR.tsv \
  --allelic-counts sandbox3/A...TTM1.allelicCounts.tsv \
  --normal-allelic-counts sandbox3/...NB1.allelicCounts.tsv \
  --output sandbox3/sandbox \
  --output-prefix
```

sandbox3/sandbox

```
(main) [mcollins@compute02 sandbox]$ ls
...TTM1_clean.af.igv.seg      ...TTM1_clean.modelBegin.af.param
...TTM1_clean.called.igv.seg  ...TTM1_clean.modelBegin.cr.param
...TTM1_clean.called.seg      ...TTM1_clean.modelBegin.seg
...TTM1_clean.cr.igv.seg      ...TTM1_clean.modelFinal.af.param
...TTM1_clean.cr.seg          ...TTM1_clean.modelFinal.cr.param
...TTM1_clean.hets.normal.tsv ...TTM1_clean.modelFinal.seg
...TTM1_clean.hets.tsv
```

```
gatk CallCopyRatioSegments \
  --I sandbox3/sandbox/...TTM1_clean.cr.seg \
  --O sandbox3/sandbox/...TTM1_clean.called.seg
```



## GATK on Real Files

```
gatk PreprocessIntervals -R /data4/shared/reference/hs38DH/hs38DH.fa --bin-length 0 --interval-merging-rule OVERLAPPING_ONLY -O
sandbox3/targets_C.preprocessed.interval_list
```

#READ COUNTS FOR ...-NB1

```
gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-...-NB1.bam -L sandbox3/targets_C.preprocessed.interval_list
--interval-merging-rule OVERLAPPING_ONLY -O sandbox3/Anorm.counts.hdf5
```

#READ COUNTS FOR ...-TTM1.bam

```
gatk CollectReadCounts -I /data4/shared/data_files/---/bams/...-...-TTM1.bam -L sandbox3/targets_C.preprocessed.interval_list
--interval-merging-rule OVERLAPPING_ONLY -O sandbox3/...TTM1.counts.hdf5
```

#PoN MADE FROM ...-NB1

```
gatk --java-options "-Xmx6500m" CreateReadCountPanelOfNormals -I sandbox3/Anorm.counts.hdf5 --minimum-interval-median-percentile 5.0 -O
sandbox3/ANpon.pon.hdf5
```

#Metastatic tumour allelic counts ...-TTM1

```
gatk --java-options "-Xmx15g" CollectAllelicCounts \
  -L /data4/shared/reference/bed_files/---.Exome_2019.bed \
  -I /data4/shared/data_files/---/bams/...-...-TTM1.bam \
  -R /data4/shared/reference/hs38DH/hs38DH.fa \
  -O A...TTM1.allelicCounts.tsv
```

#Normal blood allelic counts ...-NB1

```
gatk --java-options "-Xmx15g" CollectAllelicCounts \
  -L /data4/shared/reference/bed_files/---.Exome_2019.bed \
  -I /data4/shared/data_files/---/bams/...-...-NB1.bam \
  -R /data4/shared/reference/hs38DH/hs38DH.fa \
  -O ...NB1.allelicCounts.tsv
```

```
gatk --java-options "-Xmx12g" DenoiseReadCounts \
  -I sandbox3/...TTM1.counts.hdf5 \
  --count-panel-of-normals sandbox3/ANpon.pon.hdf5 \
  --standardized-copy-ratios sandbox3/...TTM1.standardizedCR.tsv \
  --denoised-copy-ratios sandbox3/...TTM1.denoisedCR.tsv
```

```
gatk --java-options "-Xmx31g" ModelSegments \  
  --denoised-copy-ratios sandbox3/...TTM1.denoisedCR.tsv \  
  --allelic-counts sandbox3/A...TTM1.allelicCounts.tsv \  
  --normal-allelic-counts sandbox3/...NB1.allelicCounts.tsv \  
  --output sandbox3/sandbox \  
  --output-prefix
```

sandbox3/sandbox

```
(main) [mcollins@compute02 sandbox]$ ls  
...TTM1_clean.af.igv.seg      ...TTM1_clean.modelBegin.af.param  
...TTM1_clean.called.igv.seg  ...TTM1_clean.modelBegin.cr.param  
...TTM1_clean.called.seg      ...TTM1_clean.modelBegin.seg  
...TTM1_clean.cr.igv.seg      ...TTM1_clean.modelFinal.af.param  
...TTM1_clean.cr.seg          ...TTM1_clean.modelFinal.cr.param  
...TTM1_clean.hets.normal.tsv  ...TTM1_clean.modelFinal.seg  
...TTM1_clean.hets.tsv
```

```
gatk CallCopyRatioSegments \  
  --I sandbox3/sandbox/...TTM1_clean.cr.seg \  
  --O sandbox3/sandbox/...TTM1_clean.called.seg  
  Shared data folder project maia
```

GATK4 and nextflow

<https://github.com/nf-core/sarek/tree/master/modules/nf-core/gatk4>

<https://github.com/lifebit-ai/GATK/blob/master/main.nf>

## Snakemake Tutorial

Use snakemake to streamline CNV analysis. Have it running and completed in 10 minutes.

Figure out:

- How to include a checkpoint
  - If there is a given interval list continue to CollectReadCounts
  - If not run PreProcessIntervals
  - *Run read counts, if you don't have an interval list then run preprocess intervals?*

```
rule CollectReadCounts:
rule PreProcessIntervals:
rule CreateReadCountPanelOfNormals:
```
- How to run multiple functions in parallel

Snakemake <https://snakemake-wrappers.readthedocs.io/en/stable/wrappers/bio/gatk/collectreadcounts.html>

snakemake --use-conda

### CollectReadCounts

```
rule collectreadcounts:
    input:
        bam=["mapped/a.bam"],
        intervals=["a.interval_list"],
    output:
        counts="a.counts.hdf5",
    log:
        "logs/gatk/collectreadcounts.log",
    params:
        extra="", # optional
        java_opts="", # optional
    resources:
        mem_mb=1024,
```

```
wrapper:  
  "v7.6.0/bio/gatk/collectreadcounts"
```

## ModelSegments

```
rule modelsegments_denoise_input:  
  input:  
    denoised_copy_ratios="a.denoisedCR.tsv",  
  output:  
    "a.den.modelFinal.seg",  
    "a.n.cr.seg",  
  log:  
    "logs/gatk/modelsegments_denoise.log",  
  params:  
    #prefix="a.den.test",  
    extra="", # optional  
    java_opts="", # optional  
  resources:  
    mem_mb=1024,  
  wrapper:  
    "v7.6.0/bio/gatk/modelsegments"
```

## GATK on lugh using SLURM to solve memory error

```
gatk --java-options "-Xmx12g" DenoiseReadCounts \  
  -I sandbox3/Anorm.counts.hdf5 \  
  --count-panel-of-normals sandbox3/ANpon.pon.hdf5 \  
  --standardized-copy-ratios sandbox3/Anorm.standardizedCR.tsv \  
  --denoised-copy-ratios sandbox3/Anorm.denoisedCR.tsv
```

```
gatk --java-options "-Xmx31g" ModelSegments \  
  --denoised-copy-ratios sandbox3/...TTM1.denoisedCR.tsv \  
  --allelic-counts sandbox3/A...TTM1.allelicCounts.tsv \  
  --normal-allelic-counts sandbox3/...NB1.allelicCounts.tsv \  
  --output sandbox3/sandbox \  
  --output-prefix
```

[18 March 2026, 13:48:17 GMT] org.broadinstitute.hellbender.tools.copynumber.ModelSegments done. Elapsed time: 4.74 minutes.

```
gatk CallCopyRatioSegments \  
  --I sandbox3/sandbox/Anormal.cr.seg \  
  --O sandbox3/sandbox/Anormal.called.seg
```

```
cp 'outputs' /data4/shared/projects/Maia
```

First Snakemake session

Objective for workflow: Needs to run in less than 8 hours

Input of rule all is the output of all the commands

Shell batch command

Script a python script

Conda environments

Config

Have a command at the beginning that prepares the work space and makes all the folders that are needed

If WES = padding

If WGS = bin

Things to include in thesis background from Medical Genomics II lecture 5

Above 50% due to copy number variants

Timing of copy number gains

Mutation before the copy or copy before the mutation

Large circle= copy number

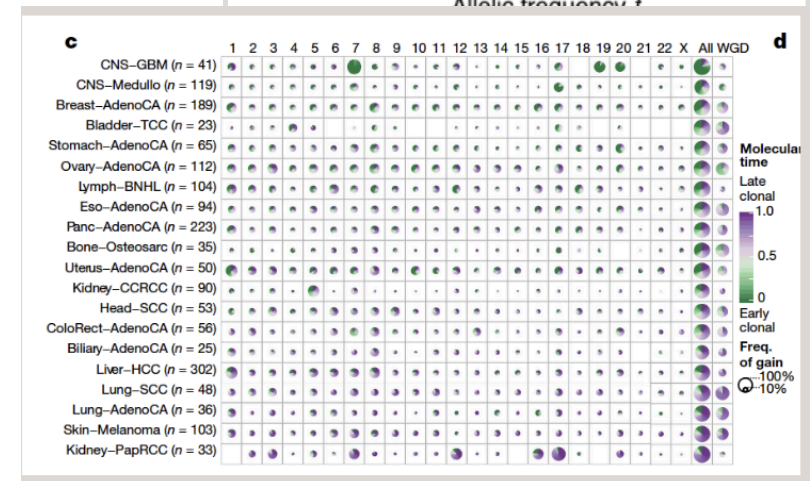
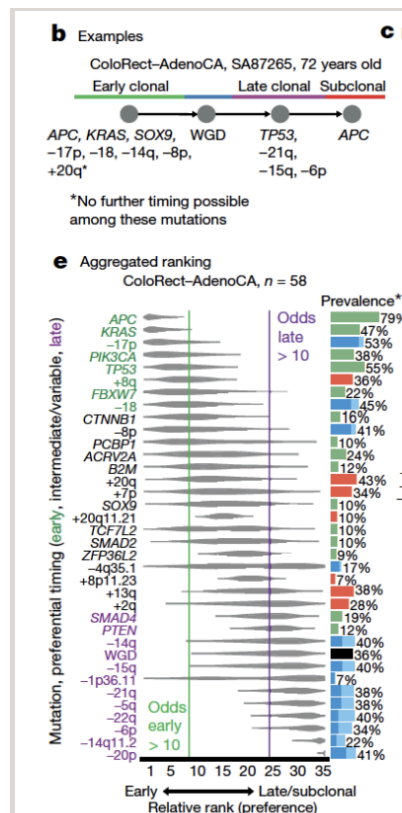
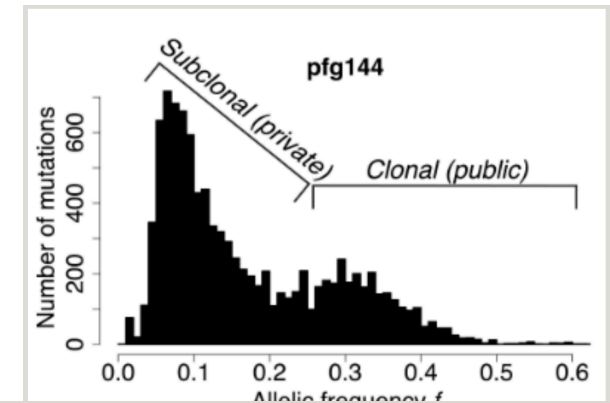
Small dots = Not a large effect

Colour = timing

- Green = early
- Purple = later on e.g. in kidney

Ranked early tumor formation

Ranked late treatment resistance and metastasis



## Snakemake Practice

<https://snakemake-wrappers.readthedocs.io/en/stable/wrappers/bio/gatk.html>

Refer to elements of the rule within the shell command string by putting them in curly braces ({}).

## Snakemake.smk

```
(snakemake-env) maia@mac snakemake % conda env export > gatksm.yml
```

```
(snakemake-env) maia@mac snakemake % ls
```

```
gatksm.yml      rules      Snakemake.smk
```

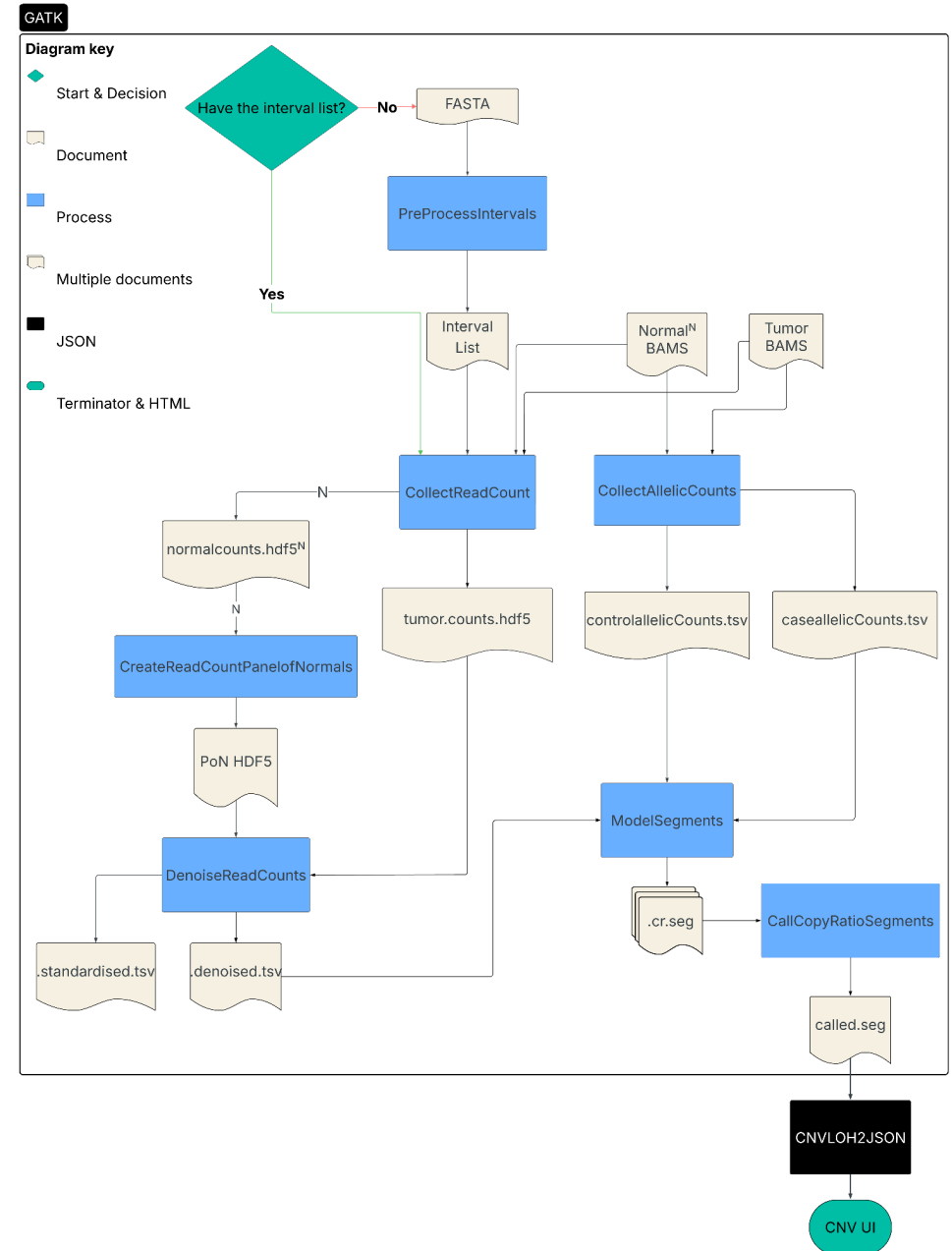
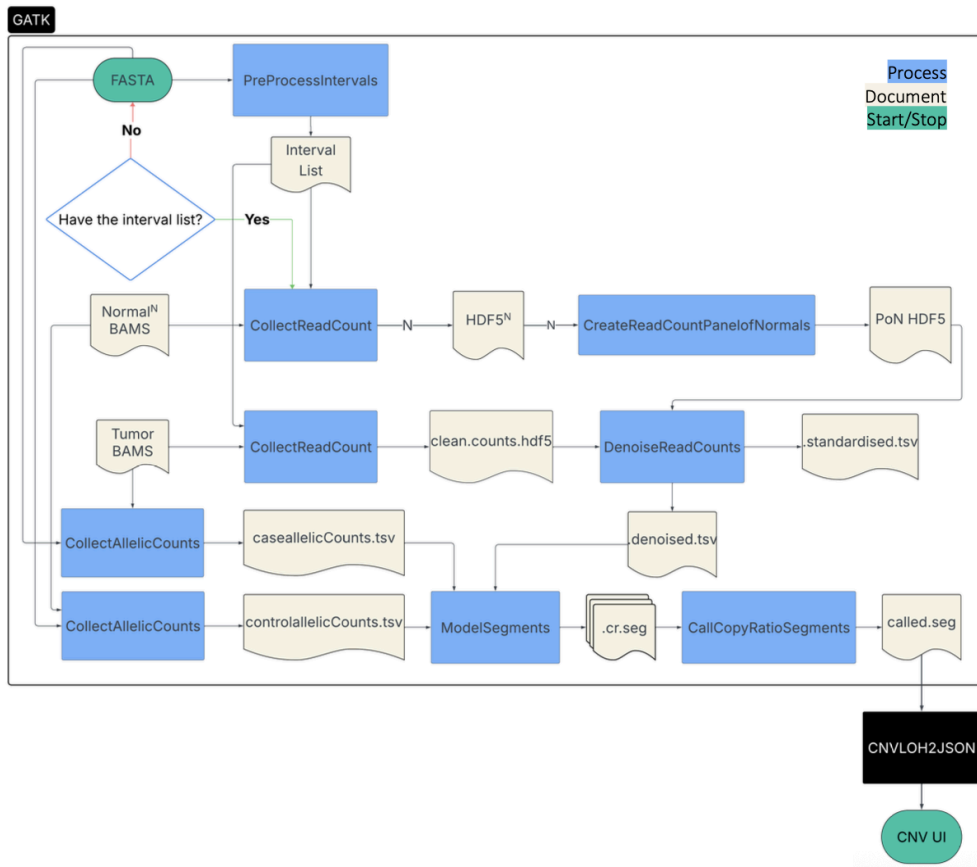


## Updated workflow flowchart

Notes from Lars following the poster presentation:

Unclear where the start is and how to get to some processes

Made the start the interval list decision and from that decision the rest of the processes can begin



## Example of a Snakemake shell

GATK collectreadcounts

```
shell(  
    "gatk --java-options '{java_opts}' CollectAllelicCounts"  
    " -I {snakemake.input.bam}"  
    " -L {snakemake.input.intervals}"  
    " -R {snakemake.input.ref}"  
    " {extra}"  
    " --tmp-dir {tmpdir}"  
    " --output {snakemake.output.counts}"  
    " {log}"  
  
)
```

## Meeting Wed 13th May

### Notes:

- Benchmarking using MCI samples
  - How accurate was the workflow?
- Include a focal detection plug in if we have time

```
cp ...TTM1_clean.hets.normal.tsv /data4/shared/projects/Maia  
cp ...NB1.allelicCounts.tsv /data4/shared/projects/Maia  
cp A...TTM1.allelicCounts.tsv /data4/shared/projects/Maia
```

Slurm\_template.sh with commands

What file from the GATK pipeline is needed to create the CNVLOH2JSON?

The command for CNVLOH2JSON is:

```
python3 /data4/shared/bin/cnv-ui/scripts \
--sample-name ...-...-TTM1 \
--sample-type WES \
--case-name ...-... \
--hets-file ...TTM1_clean.hets.tsv \
--counts-file ...TTM1.denoisedCR.tsv \
--modelsegs-file ...TTM1_clean.modelFinal.seg \
--cnvlohd-dir cnv-ui_out
```

EXCEPT, for the `--counts-file` parameter, the correct input file is not `*.denoisedCR.tsv`. That file contains basically the precursor of the `modelFinal.seg` file, and so contains estimated copy number areas that are mostly chromosome-sized.

There should be another file with counts of tiny segments (100-1000bp) that will contain many thousands of lines, which would be the precursor of this file after running `CollectReadCounts`. That needs to go into the CNVLOH2JSON to populate the data used to make all the blue dots on the plot (which are then used to also call our focal events). Could you figure out what file that is, copy it into the project folder, and then give this command a try?

Let me know if it seems like it works, and then after that you should be able to generate plots.

Requires the read counts file in TSV format

## Snakemake notes from CompBio Skills Seminar UC Berkley

```
.
├── config
│   └── config.yml
├── data
├── envs
│   └── gatksm.yml
├── logs
├── output
├── rules
│   └── Snakemake.smk
├── README.md
├── slurm.yaml
└── Snakefile
```

7 directories, 6 files

Different reference genomes, different config files - to manage different annotations associated with different reference genomes

```
snakemake --pj{n} --use-conda {file path here}
```

n = number of cores available

dry run = a rehearsal, to debug

```
.
├── config
│   └── config.yaml
├── data
│   ├── bams
│   └── reference
├── envs
│   └── gatk.yml
├── logs
├── output
├── rules
│   ├── Collectalleliccounts.smk
│   ├── Collectreadcounts.smk
│   ├── Denoisereadcounts.smk
│   ├── Modelandcall.smk
│   └── Preprocess.smk
├── README.md
├── slurm.yaml
└── Snakefile
```

9 directories, 10 files

.preprocessed.interval\_list

## Genome reference is a fasta file .fa

```
sftp> cd /data4/shared/reference/hs38DH
```

```
sftp> ls
```

```
biallelic  hs38DH.dict  hs38DH.fa  hs38DH.fa.fai
```

```
sftp> get hs38DH.fa
```

```
Fetching
```

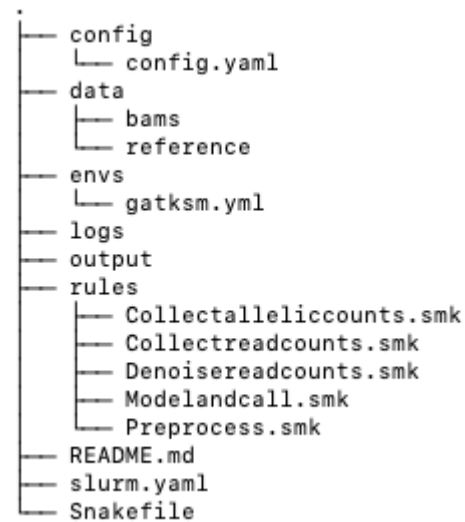
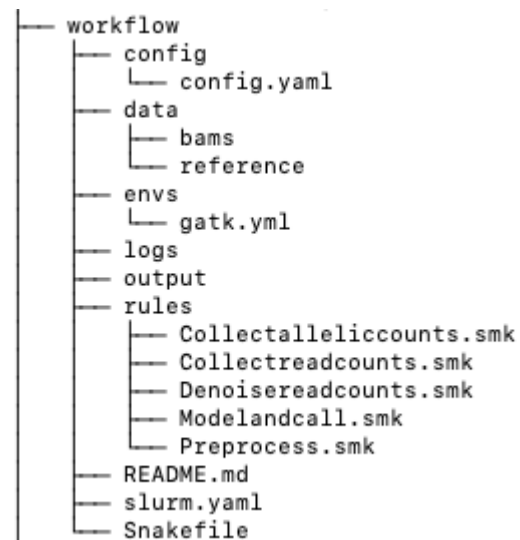
## Intervals

output/.preprocessed.interval\_list

## BAMS

bams/...-...-NB1.bam

bams/...-...-TTM1.bam



## Meeting objectives

How to distinguish between normal and tumour samples

- A folder for normal and a folder for tumor samples

One sample at a time bam = "data/bams/normal/{sample}.bam"

All samples together = expand("data/bams/normal/{sample}.bam")

Made directories manually for where to put reference files, bam files and where the output files will go.

Config file

I could specify files in the config file, other users can then edit this or I could use wildcards and users can then place their files in the directory and run it.

Will I make a rule which creates the sequence dictionary and index for the fasta file?

```
#normal_read_counts = glob_wildcards("output/{sample}.normal_counts.hdf5")
#tumor_read_counts = glob_wildcards("output/{sample}.tumor_counts.hdf5")
#std_cr = glob_wildcards("output/{sample}.standardizedCR.tsv")
#denoised_cr = glob_wildcards("output/{sample}.denoisedCR.tsv")
#normal_allelic_counts = glob_wildcards("output/{sample}.normal_alleliccounts.tsv")
#tumor_allelic_counts = glob_wildcards("output/{sample}.tumor_alleliccounts.tsv")
```

Meeting May 20th

sftp mcollins@lugh.nuigalway.ie

```
sftp> get ...
```

Start with introduction and background

Indexed bam files

Focal vs broad

SAMtools -s

```
samtools view -s 0.001 ...-NB1.bam > subsample/...-NB1-1pc.bam
```

Pc as percent

Create a small file from the bam to practice on

Index the new file

How to do subsampling

First time running snakemake

```
(base) maia@Maia-MacBook-Air workflow % snakemake --core 1 --verbose
```

Assuming unrestricted shared filesystem usage.

host: Maia-MacBook-Air.local

Building DAG of jobs...

Need to rerun job collectreadcounts\_normal because of missing output required by createreadcountpanelofnormals.

Need to rerun job denoisereadcounts because job createreadcountpanelofnormals has to be rerun.  
 Need to rerun job all because job createreadcountpanelofnormals has to be rerun.  
 Need to rerun job createreadcountpanelofnormals because job collectreadcounts\_normal has to be rerun.  
 Need to rerun job all because job collectreadcounts\_normal has to be rerun.  
 Need to rerun job collectreadcounts\_tumor because of missing output required by denoisereadcounts.  
 Need to rerun job modelsegments\_denoise\_input because job denoisereadcounts has to be rerun.  
 Need to rerun job all because job denoisereadcounts has to be rerun.  
 Need to rerun job collectalleliccounts\_normal because of missing output required by all.  
 Need to rerun job collectalleliccounts\_tumor because of missing output required by all.  
 Need to rerun job call\_copy\_ratio\_segments because of missing output required by all.  
 Need to rerun job denoisereadcounts because job collectreadcounts\_tumor has to be rerun.  
 Need to rerun job all because job collectreadcounts\_tumor has to be rerun.  
 Need to rerun job call\_copy\_ratio\_segments because job modelsegments\_denoise\_input has to be rerun.  
 Need to rerun job all because job modelsegments\_denoise\_input has to be rerun.  
 Need to rerun job all because job collectalleliccounts\_normal has to be rerun.  
 Need to rerun job all because job collectalleliccounts\_tumor has to be rerun.  
 Need to rerun job all because job call\_copy\_ratio\_segments has to be rerun.  
 output/...-TTM1.tumor\_counts.hdf5: True 0  
 output/...-TTM1.standardizedCR.tsv output/...-TTM1.denoisedCR.tsv: False 2  
 output/...-TTM1.tumor\_alleliccounts.tsv: True 0  
 output/...-TTM1.den.modelFinal.seg output/...-TTM1.n.cr.seg: False 1  
 : False 8  
 output/...-NB1.normal\_counts.hdf5: True 0  
 output/pon.hdf5: False 1  
 output/...-NB1.normal\_alleliccounts.tsv: True 0  
 output/...-TTM1.called.seg output/...-TTM1.called.igv.seg: False 1  
 shared\_storage\_local\_copies: True  
 remote\_exec: False  
 Submitting maximum 100 job(s) over 1.0 second(s).  
 Using shell: /bin/bash  
 Provided cores: 1 (use --cores to define parallelism)  
 Rules claiming more threads will be scaled down.  
 Conda environments: ignored  
 Job stats:

| job                         | count |
|-----------------------------|-------|
| collectalleliccounts_normal | 1     |



```
collectalleliccounts_tumor      1
collectreadcounts_normal        1
collectreadcounts_tumor         1
createreadcountpanelofnormals   1
denoisereadcounts               1
modelsegments_denoise_input     1
call_copy_ratio_segments        1
all                             1
total                           9
```

Resources before job selection: {'\_cores': 1, '\_nodes': 9223372036854775807, '\_job\_count': 9223372036854775807}

Ready jobs: 4

Select jobs to execute...

Selected jobs: 1

Resources after job selection: {'\_cores': 0, '\_nodes': 9223372036854775806, '\_job\_count': 100}

Execute 1 jobs...

[Wed May 20 16:46:00 2026]

localrule collectreadcounts\_tumor:

input: data/bams/tumor/...-...-TTM1.bam, output/preprocessedintervals.list

output: output/...-...-TTM1.tumor\_counts.hdf5

log: logs/gatk/...-...-TTM1.collectreadcountstumor.log

jobid: 3

reason: Missing output files: output/...-...-TTM1.tumor\_counts.hdf5

wildcards: sample=...-...-TTM1

resources: tmpdir=/var/folders/js/47lf8p8j62932tb9qj2cpp7r0000gn/T, mem\_mb=1024, mem=1.02 GB, mem\_mib=977

Waiting for running jobs to complete.

Using GATK jar /Users/maia/miniconda3/share/gatk4-4.6.2.0-0/gatk-package-4.6.2.0-local.jar

Running:

```
java -Dsamjdk.use_async_io_read_samtools=false -Dsamjdk.use_async_io_write_samtools=true -Dsamjdk.use_async_io_write_tribble=false -Dsamjdk.compression_level=2 -jar
/Users/maia/miniconda3/share/gatk4-4.6.2.0-0/gatk-package-4.6.2.0-local.jar CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule
OVERLAPPING_ONLY -O output/...-...-TTM1.tumor_counts.hdf5
```

Resources before job selection: {'\_cores': 0, '\_nodes': 9223372036854775806, '\_job\_count': 100}

Ready jobs: 3

Select jobs to execute...

Selected jobs: 0

Resources after job selection: {'\_cores': 0, '\_nodes': 9223372036854775806, '\_job\_count': 100}

Waiting for running jobs to complete.

16:46:02.340 INFO NativeLibraryLoader - Loading libgkl\_compression.dylib from  
jar:file:/Users/maia/miniconda3/share/gatk4-4.6.2.0-0/gatk-package-4.6.2.0-local.jar!/com/intel/gkl/native/libgkl\_compression.dylib

Resources before job selection: {'\_cores': 0, '\_nodes': 9223372036854775806, '\_job\_count': 100}

Ready jobs: 3

Selected jobs: 0

Resources after job selection: {'\_cores': 0, '\_nodes': 9223372036854775806, '\_job\_count': 100}

Waiting for running jobs to complete.

16:46:02.812 INFO CollectReadCounts - -----

16:46:02.814 INFO CollectReadCounts - The Genome Analysis Toolkit (GATK) v4.6.2.0

16:46:02.815 INFO CollectReadCounts - For support and documentation go to <https://software.broadinstitute.org/gatk/>

16:46:02.815 INFO CollectReadCounts - Executing as maia@Maia-MacBook-Air.local on Mac OS X v15.6 x86\_64

16:46:02.815 INFO CollectReadCounts - Java runtime: OpenJDK 64-Bit Server VM v17.0.18+0

16:46:02.815 INFO CollectReadCounts - Start Date/Time: 20 May 2026 at 16:46:02 IST

16:46:02.815 INFO CollectReadCounts - -----

16:46:02.815 INFO CollectReadCounts - -----

16:46:02.817 INFO CollectReadCounts - HTSJDK Version: 4.2.0

16:46:02.817 INFO CollectReadCounts - Picard Version: 3.4.0

16:46:02.817 INFO CollectReadCounts - Built for Spark Version: 3.5.0

16:46:02.825 INFO CollectReadCounts - HTSJDK Defaults.COMPRESSION\_LEVEL : 2

16:46:02.825 INFO CollectReadCounts - HTSJDK Defaults.USE\_ASYNC\_IO\_READ\_FOR\_SAMTOOLS : false

16:46:02.825 INFO CollectReadCounts - HTSJDK Defaults.USE\_ASYNC\_IO\_WRITE\_FOR\_SAMTOOLS : true

16:46:02.825 INFO CollectReadCounts - HTSJDK Defaults.USE\_ASYNC\_IO\_WRITE\_FOR\_TRIBBLE : false

16:46:02.826 INFO CollectReadCounts - Deflater: IntelDeflater

16:46:02.826 INFO CollectReadCounts - Inflater: IntelInflater

16:46:02.826 INFO CollectReadCounts - GCS max retries/reopens: 20

16:46:02.826 INFO CollectReadCounts - Requester pays: disabled

16:46:02.826 INFO CollectReadCounts - Initializing engine

16:46:03.019 INFO CollectReadCounts - Shutting down engine

[20 May 2026 at 16:46:03 IST] org.broadinstitute.hellbender.tools.copynumber.CollectReadCounts done. Elapsed time: 0.01 minutes.

Runtime.totalMemory()=260046848

\*\*\*\*\*

A USER ERROR has occurred: Badly formed genome unclippedLoc: Query interval "@HVN:1.6" is not valid for this input.

\*\*\*\*\*

Set the system property GATK\_STACKTRACE\_ON\_USER\_EXCEPTION (--java-options '-DGATK\_STACKTRACE\_ON\_USER\_EXCEPTION=true') to print the stack trace.

Full Traceback (most recent call last):

File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/executors/local.py", line 432, in run\_wrapper  
run(

```

~~~~~
    input,
    ^^^^^^
...<23 lines>...
    runtime_paths,
    ^^^^^^^^^^^^^^^^^
)
^
File "/Users/maia/Documents/workflow/rules/Collectreadcounts.smk", line 93, in __rule_collectreadcounts_tumor
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/shell.py", line 359, in __new__
    raise sp.CalledProcessError(retcode, cmd)
subprocess.CalledProcessError: Command 'set -euo pipefail; gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule
OVERLAPPING_ONLY -O output/...-...-TTM1.tumor_counts.hdf5' returned non-zero exit status 2.
Full Traceback (most recent call last):
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/executors/local.py", line 432, in run_wrapper
    run(
~~~~~
    input,
    ^^^^^^
...<23 lines>...
    runtime_paths,
    ^^^^^^^^^^^^^^^^^
)
^
File "/Users/maia/Documents/workflow/rules/Collectreadcounts.smk", line 93, in __rule_collectreadcounts_tumor
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/shell.py", line 359, in __new__
    raise sp.CalledProcessError(retcode, cmd)
subprocess.CalledProcessError: Command 'set -euo pipefail; gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule
OVERLAPPING_ONLY -O output/...-...-TTM1.tumor_counts.hdf5' returned non-zero exit status 2.
During handling of the above exception, another exception occurred:
Traceback (most recent call last):
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/executors/local.py", line 267, in _callback
    raise ex
File "/Users/maia/miniconda3/lib/python3.13/concurrent/futures/thread.py", line 59, in run
    result = self.fn(*self.args, **self.kwargs)
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/executors/local.py", line 236, in cached_or_run
    self.workflow.async_run(self.acached_or_run(job, run_func, *args))

```

```

~~~~~AAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAAA
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/workflow.py", line 268, in async_run
    return runner.run(coro)
~~~~~AAAAAA
File "/Users/maia/miniconda3/lib/python3.13/asyncio/runners.py", line 118, in run
    return self._loop.run_until_complete(task)
~~~~~AAAAAA
File "/Users/maia/miniconda3/lib/python3.13/asyncio/base_events.py", line 725, in run_until_complete
    return future.result()
~~~~~AA
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/executors/local.py", line 253, in acached_or_run
    run_func(*args)
~~~~~AAAAAA
File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/executors/local.py", line 469, in run_wrapper
    raise RuleException(
...<3 lines>...
)
snakemake.exceptions.RuleException: CalledProcessError in file "/Users/maia/Documents/workflow/rules/Collectreadcounts.smk", line 41:
Command 'set -euo pipefail; gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING_ONLY -O
output/...-...-TTM1.tumor_counts.hdf5' returned non-zero exit status 2.
RuleException:
CalledProcessError in file "/Users/maia/Documents/workflow/rules/Collectreadcounts.smk", line 41:
Command 'set -euo pipefail; gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING_ONLY -O
output/...-...-TTM1.tumor_counts.hdf5' returned non-zero exit status 2.
[Wed May 20 16:46:03 2026]
Error in rule collectreadcounts_tumor:
  message: None
  jobid: 3
  input: data/bams/tumor/...-...-TTM1.bam, output/preprocessedintervals.list
  output: output/...-...-TTM1.tumor_counts.hdf5
  log: logs/gatk/...-...-TTM1.collectreadcountstumor.log (check log file(s) for error details)
  shell:
    gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING_ONLY -O output/...-...-TTM1.tumor_counts.hdf5
    (command exited with non-zero exit code)
Shutting down, this might take some time.
Exiting because a job execution failed. Look below for error messages
[Wed May 20 16:46:03 2026]

```

Error in rule collectreadcounts\_tumor:

message: None

jobid: 3

input: data/bams/tumor/...-...-TTM1.bam, output/preprocessedintervals.list

output: output/...-...-TTM1.tumor\_counts.hdf5

log: logs/gatk/...-...-TTM1.collectreadcountstumor.log (check log file(s) for error details)

shell:

```
gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING_ONLY -O output/...-...-TTM1.tumor_counts.hdf5
(command exited with non-zero exit code)
```

Complete log(s): /Users/maia/Documents/workflow/.snakemake/log/2026-05-20T164600.420224.snakemake.log

unlocking

Full Traceback (most recent call last):

File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/cli.py", line 2308, in args\_to\_api

dag\_api.execute\_workflow(

~~~~~^

executor=args.executor,

^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

...<46 lines>...

scheduler\_settings=scheduler\_settings,

^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

)

^

File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/api.py", line 646, in execute\_workflow

workflow.execute(

~~~~~^

executor\_plugin=executor\_plugin,

^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

...<4 lines>...

updated\_files=updated\_files,

^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^

)

^

File "/Users/maia/miniconda3/lib/python3.13/site-packages/snakemake/workflow.py", line 1507, in execute

raise WorkflowError("At least one job did not complete successfully.")

snakemake\_interface\_common.exceptions.WorkflowError: At least one job did not complete successfully.

WorkflowError:

At least one job did not complete successfully.

## Creating subsamples

```
AFE4-TTP1-Sample.txt ...-AFE4-TTM2.bam.bai ...-AFE4-TTM6.bam ...-...-TTM1.bam.bai
AFE-NT1-Sample.txt ...-AFE4-TTM3.bam ...-AFE4-TTM6.bam.bai ...-...-TTP1.bam
...-AFE4-NT1.bam ...-AFE4-TTM3.bam.bai ...-AFE4-TTP1.bam ...-...-TTP1.bam.bai
...-AFE4-NT1.bam.bai ...-AFE4-TTM4.bam ...-AFE4-TTP1.bam.bai subsample
...-AFE4-TTM1.bam ...-AFE4-TTM4.bam.bai ...-...-NB1.bam
...-AFE4-TTM1.bam.bai ...-AFE4-TTM5.bam ...-...-NB1.bam.bai
...-AFE4-TTM2.bam ...-AFE4-TTM5.bam.bai ...-...-TTM1.bam
```

```
samtools view -s 0.001 ...-AFE4-NT1.bam > subsample/...-AFE4-NT1-1pc.bam
samtools view -s 0.001 ...-AFE4-TTM1.bam > subsample/...-AFE4-TTM1-1pc.bam
```

## Error with tumor read counts in Snakemake

### RuleException:

CalledProcessError in file "/Users/maia/Documents/workflow/rules/Collectreadcounts.smk", line 41:

Command 'set -euo pipefail; gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING\_ONLY -O output/...-...-TTM1.tumor\_counts.hdf5' returned non-zero exit status 2.

[Mon May 25 20:43:45 2026]

Error in rule collectreadcounts\_tumor:

message: None

jobid: 3

input: data/bams/tumor/...-...-TTM1.bam, output/preprocessedintervals.list

output: output/...-...-TTM1.tumor\_counts.hdf5

log: logs/gatk/...-...-TTM1.collectreadcountstumor.log (check log file(s) for error details)

shell:

gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING\_ONLY -O

output/...-...-TTM1.tumor\_counts.hdf5

(command exited with non-zero exit code)

Shutting down, this might take some time.

Exiting because a job execution failed. Look below for error messages

[Mon May 25 20:43:45 2026]

Error in rule collectreadcounts\_tumor:

message: None

jobid: 3

input: data/bams/tumor/...-...-TTM1.bam, output/preprocessedintervals.list

output: output/...-...-TTM1.tumor\_counts.hdf5

log: logs/gatk/...-...-TTM1.collectreadcountstumor.log (check log file(s) for error details)

shell:

```
gatk CollectReadCounts -I data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING_ONLY -O
output/...-...-TTM1.tumor_counts.hdf5
```

(command exited with non-zero exit code)

Complete log(s): /Users/maia/Documents/workflow/.snakemake/log/2026-05-25T204342.498836.snakemake.log

RuleException:

CalledProcessError in file "/Users/maia/Documents/workflow/rules/Collectreadcounts.smk", line 37:

Command 'set -euo pipefail; /Users/maia/miniconda3/bin/python3.13 /Users/maia/Documents/workflow/.snakemake/scripts/tmproi0dool.wrapper.py' returned non-zero exit status 1.

[Mon May 25 20:56:22 2026]

Error in rule collectreadcounts\_tumor:

message: None

jobid: 3

input: data/bams/tumor/NA.bam, output/preprocessedintervals.list

output: output/NA.tumor\_counts.hdf5

log: logs/gatk/NA.collectreadcountstumor.log (check log file(s) for error details)

Shutting down, this might take some time.

(base) maia@Mac gatk % cat ...-...-TTM1.collectreadcountstumor.log

Using GATK jar /Users/maia/miniconda3/share/gatk4-4.6.2.0-0/gatk-package-4.6.2.0-local.jar

Running:

```
java -Dsamjdk.use_async_io_read_samtools=false -Dsamjdk.use_async_io_write_samtools=true -Dsamjdk.use_async_io_write_tribble=false -Dsamjdk.compression_level=2
-Xmx819M -Djava.io.tmpdir=/var/folders/js/47lf8p8j62932tb9qj2cpp7r0000gn/T -jar /Users/maia/miniconda3/share/gatk4-4.6.2.0-0/gatk-package-4.6.2.0-local.jar CollectReadCounts -I
data/bams/tumor/...-...-TTM1.bam -L output/preprocessedintervals.list --interval-merging-rule OVERLAPPING_ONLY --tmp-dir
/var/folders/js/47lf8p8j62932tb9qj2cpp7r0000gn/T/tmpdkzd_keg --output output/...-...-TTM1.tumor_counts.hdf5
21:13:54.150 INFO NativeLibraryLoader - Loading libgkl_compression.dylib from
jar:file:/Users/maia/miniconda3/share/gatk4-4.6.2.0-0/gatk-package-4.6.2.0-local.jar!/com/intel/gkl/native/libgkl_compression.dylib
21:13:54.464 INFO CollectReadCounts - -----
```

```

21:13:54.467 INFO CollectReadCounts - The Genome Analysis Toolkit (GATK) v4.6.2.0
21:13:54.467 INFO CollectReadCounts - For support and documentation go to https://software.broadinstitute.org/gatk/
21:13:54.467 INFO CollectReadCounts - Executing as maia@Mac.fritz.box on Mac OS X v15.6 x86_64
21:13:54.467 INFO CollectReadCounts - Java runtime: OpenJDK 64-Bit Server VM v17.0.18+0
21:13:54.467 INFO CollectReadCounts - Start Date/Time: 25 May 2026 at 21:13:54 IST
21:13:54.467 INFO CollectReadCounts - -----
21:13:54.467 INFO CollectReadCounts - -----
21:13:54.469 INFO CollectReadCounts - HTSJDK Version: 4.2.0
21:13:54.470 INFO CollectReadCounts - Picard Version: 3.4.0
21:13:54.470 INFO CollectReadCounts - Built for Spark Version: 3.5.0
21:13:54.475 INFO CollectReadCounts - HTSJDK Defaults.COMPRESSION_LEVEL : 2
21:13:54.475 INFO CollectReadCounts - HTSJDK Defaults.USE_ASYNC_IO_READ_FOR_SAMTOOLS : false
21:13:54.475 INFO CollectReadCounts - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_SAMTOOLS : true
21:13:54.475 INFO CollectReadCounts - HTSJDK Defaults.USE_ASYNC_IO_WRITE_FOR_TRIBBLE : false
21:13:54.475 INFO CollectReadCounts - Deflater: IntelDeflater
21:13:54.475 INFO CollectReadCounts - Inflater: IntelInflater
21:13:54.475 INFO CollectReadCounts - GCS max retries/reopens: 20
21:13:54.475 INFO CollectReadCounts - Requester pays: disabled
21:13:54.476 INFO CollectReadCounts - Initializing engine
21:13:54.749 INFO CollectReadCounts - Shutting down engine
[25 May 2026 at 21:13:54 IST] org.broadinstitute.hellbender.tools.copynumber.CollectReadCounts done. Elapsed time: 0.01 minutes.
Runtime.totalMemory()=225443840

```

\*\*\*\*\*  
A USER ERROR has occurred: Badly formed genome unclippedLoc: Query interval "@HD VN:1.6" is not valid for this input.

\*\*\*\*\*  
Set the system property GATK\_STACKTRACE\_ON\_USER\_EXCEPTION (--java-options '-DGATK\_STACKTRACE\_ON\_USER\_EXCEPTION=true') to print the stack trace.

I first thought that maybe the createreadcount\_tumor is not expecting preprocessintervals file with a header @HD VN:1.6".

I viewed the targets\_C.preprocessed.interval\_list in sandbox3. Preprocessed intervals generated from GATK in lugh on ---Files

@HD VN:1.6

@SQ SN:chr1 LN:248956422 M5:6aef897c3d6ff0c78aff06ac189178dd UR:file:///data4/shared/reference/hs38DH/hs38DH.fa

Preprocessed intervals generated from Snakemake on ---Files

(base) maia@Mac output % head preprocessintervals.list

@HD VN:1.6

@SQ SN:chr1 LN:248956422 M5:6aef897c3d6ff0c78aff06ac189178dd UR:file:///data4/shared/reference/hs38DH/hs38DH.fa

Removed the -L parameter from preprocess.smk and it worked.

Have the preprocessintervals.interval\_list, the normal and tumor read counts and the pon.



May 28th

```
rule modelsegments_denoise_input:
    input:
        denoised_copy_ratio="output/{sample}.denoisedCR.tsv"
    output:
        "output/{sample}.den.modelFinal.seg",
        copy_ratio_seg="output/{sample}.n.cr.seg"
    conda:
        "envs/gatk.yml"
    log:
        "logs/gatk/modelsegments_denoise_{sample}.log"
    threads: 1
    params:
        #prefix="a.den.test",
        extra="", # optional
        java_opts="" # optional
    resources:
        mem_mb=1024
    wrapper:
        "v7.6.0/bio/gatk/modelsegments"
```

Add java\_opts\_model=config["java\_opts\_model"]

Add

allelic\_counts="output/{sample}.tumor\_alleliccounts.tsv"

normal\_allelic\_counts="output/{sample}.normal\_alleliccounts.tsv"

MissingInputException in rule collectalleliccounts\_normal in file "/Users/maia/Documents/workflow/rules/Collectalleliccounts.smk", line 3:

Missing input files for rule collectalleliccounts\_normal:

output: output/...-...-NB1.normal\_alleliccounts.tsv

wildcards: sample=...-...-NB1

affected files:

...-...-NB1

MissingInputException in rule collectalleliccounts\_normal in file "/Users/maia/Documents/workflow/rules/Collectalleliccounts.smk", line 3:

Missing input files for rule collectalleliccounts\_normal:

output: output/AU.normal\_alleliccounts.tsv

wildcards: sample=AU

affected files:

AU

Solution: Need to specify tumor normal pairs in the config file

First full run of Snakemake workflow

```
16
17 #specify your samples here
18 #tumor_samples and normal pair
19 pairs:
20 | "AUR-AG12-TTM1": "AUR-AG12-NB1"
21
```

normal\_counts = `lambda wc: f"output/normal_counts/{pairs.loc[wc.sample, 'normal']}.normal_alleliccounts.tsv"`

collectalleliccounts\_normal

Normal\_counts Processed 80794678 total loci in 103.0 minutes.

CollectAllelicCounts done. Elapsed time: 110.32 minutes.

collectalleliccounts\_tumor

Runtime.totalMemory()=12222201856

Elapsed time: 22.86 minutes.

ModelSegments done. Elapsed time: 20.42 minutes.

Waiting at most 5 seconds for missing files:

output/sandbox/.....TTM1.den.modelFinal.seg (missing locally)

output/sandbox/.....TTM1.den.cr.seg (missing locally)

MissingOutputException in rule modelsegments\_denoise\_input in file "/Users/maia/Documents/workflow/rules/Modelandcall.smk", line 2:

Job 8 completed successfully, but some output files are missing. Missing files after 5 seconds. This might be due to filesystem latency. If that is the case, consider to increase the wait time with --latency-wait:

output/sandbox/.....TTM1.den.modelFinal.seg (missing locally, parent dir contents: .....TTM1.af.igv.seg, .....TTM1.modelBegin.cr.param, .....TTM1.cr.seg, .....TTM1.modelFinal.cr.param, .....TTM1.modelBegin.seg, .....TTM1.hets.tsv, .....TTM1.cr.igv.seg, .....TTM1.modelBegin.af.param, .....TTM1.modelFinal.af.param, .....TTM1.hets.normal.tsv, .....TTM1.modelFinal.seg)

output/sandbox/.....TTM1.den.cr.seg (missing locally, parent dir contents: .....TTM1.af.igv.seg, .....TTM1.modelBegin.cr.param, .....TTM1.cr.seg, .....TTM1.modelFinal.cr.param, .....TTM1.modelBegin.seg, .....TTM1.hets.tsv, .....TTM1.cr.igv.seg, .....TTM1.modelBegin.af.param, .....TTM1.modelFinal.af.param, .....TTM1.hets.normal.tsv, .....TTM1.modelFinal.seg)

Shutting down, this might take some time.

Exiting because a job execution failed. Look below for error messages

[Tue Jun 2 20:13:38 2026]

Error in rule modelsegments\_denoise\_input:

message: None

jobid: 8

input: output/.....TTM1.denoisedCR.tsv, output/tumor\_counts/.....TTM1.tumor\_alleliccounts.tsv, output/normal\_counts/.....NB1.normal\_alleliccounts.tsv

output: output/sandbox/.....TTM1.den.modelFinal.seg, output/sandbox/.....TTM1.den.cr.seg

log: logs/gatk/modelsegments\_denoise\_.....TTM1.log (check log file(s) for error details)

shell:

gatk --java-options '-Xmx31g' ModelSegments --denoised-copy-ratios output/.....TTM1.denoisedCR.tsv --allelic-counts output/tumor\_counts/.....TTM1.tumor\_alleliccounts.tsv --normal-allelic-counts output/normal\_counts/.....NB1.normal\_alleliccounts.tsv --output output/sandbox/ --output-prefix .....TTM1

(command exited with non-zero exit code)

Complete log(s): /Users/maia/Documents/workflow/.snakemake/log/2026-06-02T195305.878174.snakemake.log

WorkflowError:

At least one job did not complete successfully.

20:31:16.065 WARN AlleleFractionInitializer - The maximum-likelihood estimate for the global parameter AF\_outlier\_probability (0.148815) was near its boundary (0.150000), the model is likely not a good fit to the data! Consider changing parameters for filtering homozygous sites.

```
.
├── config
│   └── config.yaml
├── data
│   ├── bams
│   │   ├── normal
│   │   └── tumor
│   └── reference
├── logs
├── output
│   ├── normal_counts
│   ├── sandbox
│   └── tumor_counts
├── README.md
├── rules
│   ├── Collectalleliccounts.smk
│   ├── Collectreadcounts.smk
│   ├── Denoisereadcounts.smk
│   ├── envs
│   │   └── gatk.yml
│   ├── Modelandcall.smk
│   └── Preprocess.smk
├── slurm.yaml
└── Snakefile
```

## Running Snakemake with multiple tumor samples

```
pairs:
    "...-...-TTM1": "...-...-NB1"
    "...-...-TTP1": "...-...-NB1"

PAIRS = config["pairs"]
TUMOR_SAMPLES = list(PAIRS.keys())
NORMAL_SAMPLES = list(set(PAIRS.values())) # set will make sure the same normal sample is not used multiple times for the PON

rule createreadcountpanelofnormals:
    input:
        hdf5 = expand("output/{sample}.normal_counts.hdf5", sample=NORMAL_SAMPLES)
    output:
        pon = "output/pon.hdf5"
    conda:
        "envs/gatk.yml"
    log:
        "logs/gatk/createreadcountpanelofnormals.log"
    threads: 1
    params:
```

```

flags = lambda wildcards, input: " ".join([f"-I {f}" for f in input.hdf5])

shell:

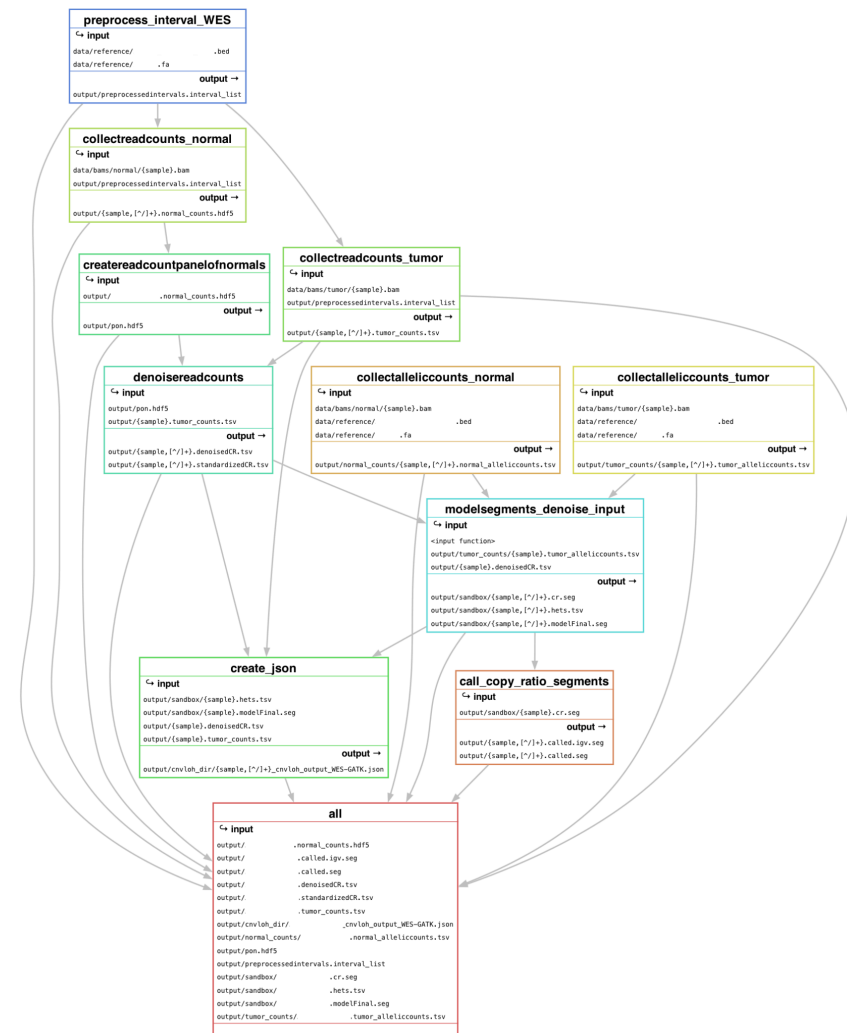
"gatk CreateReadCountPanelOfNormals "
"{params.flags} "
"-O {output.pon} "

```

## DAG flowcharts

(base) maia@Mac workflow % snakemake --forceall --dag | dot -Tpdf > dag.pdf

(base) maia@MAC-0C9E47 workflow % snakemake --forceall --filegraph | dot -Tpng > filegraph.png



Meeting June 4th

Instead of \*.hdf5 use \*.tsv for collect read counts

```
.
├── config
│   └── config.yaml
├── data
│   ├── bams
│   │   ├── normal
│   │   └── tumor
│   └── reference
├── envs
│   └── gatk.yml
├── logs
│   └── gatk
├── output
│   ├── normal_counts
│   ├── sandbox
│   └── tumor_counts
├── README.md
├── rules
│   ├── Collectalleliccounts.smk
│   ├── Collectreadcounts.smk
│   ├── Denoisereadcounts.smk
│   ├── Modelandcall.smk
│   └── Preprocess.smk
└── Snakefile
```

CNVLOH2JSON ran with the input files from Snakemake workflow

June 5th

```
`python3 /data4/shared/bin/cnv-ui/scripts/cnvloh2json.py \  
--sample-name \  
--case-name \  
--modelsegs-file ../modelFinal.seg \  
--sample-type Tumor_Normal \  
--case-name \  
--cnvloh-dir <out_dir> \  
--denoised-file.denoisedCR.tsv \  
--counts-file .tumor_counts.tsv \  
--hets-file .hets.tsv \  
--software GATK \  
--data-type WES`
```

(Replace parameters with appropriate Snakemake inputs and wildcards. Make sure the output directory provided exists.)

After that, you take the JSON created, which contains all the info from the various different GATK outputs in one file. You can then run the plotting script with:

```
`python3 /data4/shared/bin/cnv-ui/scripts/plot_interactive_cnvloh.py \  
--cnvloh-json <json created by last step> \  
--cnvloh-dir <out_dir> \  
--genome-info /data4/shared/bin/cnv-ui/data/genome-info_GRCh38.json`
```



(To get this step to run, you may need to create a conda environment with 'bokeh==2.3.3', which will force an older version of Python.)

There are a few issues. First, as indicated by the error saying your model is a poor fit for the data, some of the parameters you're running with right now are probably not great. In a healthy genome with no CNVs, we would expect there to be 23 or 24 segments in the modelSegs file, one for each chromosome; for a heavily altered tumour genome, there might be over a hundred. Currently, your analysis is outputting 8762 segments, which is causing the downstream steps to be very slow and the plotting code is struggling a bit. Using better model parameters, many of those tiny segments should become merged together, which will clear that up. I might suggest trying that before the plotting, because after letting it run for ~10 mins, the plotting still wasn't done, whereas usually it only takes a couple seconds.

The parameters my previous group settled on after extensive testing (which I think should get entered in the ModelSegments step) are:

"nocppf": 1.25

"kvaf": 0.025

"kvcr": 0.0

"ksaf": 1.0

"scitaf": 2.0

"scitcr": 2.0

Please refer to the GATK documentation, which hopefully explains what each of those things are (as I just have the parameter list and not a set of explanations).

We also still need some supporting data files to properly annotate the plots with gene information. I'll work on retrieving those, but once I have them it should be a simple matter of rerunning just the plotting step to add everything in.

Snakemake running with ModelSegment parameters and two scripts for creating html

June 6th

```
rule plot_interactive_cnvloh:
    input:
        json = "output/cnvloh_dir/{sample}.json"
    output:
        html = "output/cnvloh_dir/{sample}_interactive.html"
    conda:
        "envs/json.yml"
    log:
        "logs/json/{sample}.html.log"
    threads: 1
    resources:
        mem_mb=1024
    shell:
        "python3 py_json/plot_interactive_cnvloh.py "
        "--cnvloh-json {input.json} "
        "--cnvloh-dir output/cnvloh_dir "
        "--genome-info py_json/genome-info_GRCh38.json"
```

```
rule all:
    input:
        "output/preprocessedintervals.interval_list",
        expand("output/{sample}.normal_counts.hdf5", sample=NORMAL_SAMPLES),
        expand("output/{sample}.tumor_counts.tsv", sample=TUMOR_SAMPLES),
        "output/pon.hdf5",
        expand("output/{sample}.standardizedCR.tsv", sample=TUMOR_SAMPLES),
        expand("output/{sample}.denoisedCR.tsv", sample=TUMOR_SAMPLES),
```

```

expand("output/normal_counts/{sample}.normal_alleliccounts.tsv", sample=NORMAL_SAMPLES),
expand("output/tumor_counts/{sample}.tumor_alleliccounts.tsv", sample=TUMOR_SAMPLES),
expand("output/sandbox/{sample}.modelFinal.seg", sample=TUMOR_SAMPLES),
expand("output/sandbox/{sample}.cr.seg", sample=TUMOR_SAMPLES),
expand("output/{sample}.called.seg", sample=TUMOR_SAMPLES),
expand("output/{sample}.called.igv.seg", sample=TUMOR_SAMPLES),
expand("output/cnvloh_dir/{sample}_cnvloh_output_WES-GATK.json", sample=TUMOR_SAMPLES)

```

Traceback (most recent call last):

File `"/Users/maia/Documents/workflow/py_json/plot_interactive_cnvloh.py"`, line 1579, in `<module>`

```

plot_all_chrom_html(
~~~~~^

    genome_info=genome_info,
    ~~~~~
...<7 lines>...
    metadata=metadata_dict,
    ~~~~~
)
^

```

File `"/Users/maia/Documents/workflow/py_json/plot_interactive_cnvloh.py"`, line 1319, in `plot_all_chrom_html`

```

plot_genome_html_bokeh(
~~~~~^

    out_path=out_path,
    ~~~~~
...<7 lines>...
    metadata=metadata,
    ~~~~~
)
^

```

File `"/Users/maia/Documents/workflow/py_json/plot_interactive_cnvloh.py"`, line 1290, in `plot_genome_html_bokeh`

```

cnvloh_bokeh_plot_genome(
~~~~~^

    genes_table=genes_table,
    ~~~~~

```

...<7 lines>...

```
    male_adjustment=is_male,
    ~~~~~
)
^
```

File "/Users/maia/Documents/workflow/py\_json/plot\_interactive\_cnvloh\_bokeh.py", line 3683, in generate\_genome\_plot

```
source_seg = create_sourceseg_datasource(
```

```
    cnvloh_dna_dfs=cnvloh_dna_dfs,
```

...<5 lines>...

```
    baseline_adj_list=baseline_adj_list,
```

```
)
```

File "/Users/maia/Documents/workflow/py\_json/plot\_interactive\_cnvloh\_bokeh.py", line 322, in create\_sourceseg\_datasource

```
"cnv_num_copy": (2 ** (np.array(cnvloh_dna_dfs["seg_df"]["LOG2"]) + 1))
```

```
    ~~~~~
```

```
    / (1 + np.array(baseline_adj_list)),
```

```
    ~~~~~
```

**ValueError:** operands could not be broadcast together with shapes (9418,) (9411,)